The Effects of Surge Pricing on Driver Behavior in the Ride-Sharing Market: Evidence from a Quasi-Experiment*

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Surge pricing has been used to coordinate supply and demand in the ride-sharing industry, but its causal effects on driver behavior remain unclear. This motivates us to examine how surge pricing causally affects driver earnings and labor supply by leveraging a unique quasi-experiment, in which a leading ride-sharing company in China introduced surge pricing in two cities at different times. Using a difference-in-differences design with the *causal forest method*, we find that surge pricing led to increases in drivers' weekly revenue. Decomposing the weekly revenue into "intensive margin" and "extensive margin" factors, we discover two countervailing effects at play: a cherry-picking effect and a competition effect, and the daily revenue decreased because the latter dominated. Consequently, the increased weekly revenue can be explained by the extensive margin: drivers worked on more days to compensate for the decreased daily revenue, a result consistent with the income targeting behavior. Finally, we examine heterogeneous treatment effects across drivers, and find that surge pricing enticed more part-time drivers to flood the market and crowd out full-time drivers, and that the increase in the drivers' weekly revenue was primarily driven by part-time drivers. Therefore, the benefit of surge pricing was unevenly distributed across drivers.

Key words: Surge pricing, Ride-sharing, Labor supply, Extensive margin, Intensive margin, Cherry-picking, Competition

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1. Introduction

Online platforms in the sharing economy match independent service providers with individual customers. Such two-sided nature brings forward new challenges for developing pricing mechanisms to coordinate demand and supply. This is especially relevant for ride-sharing platforms because each service provider (driver) can flexibly decide his work schedule. Ride-sharing platforms (e.g., Uber, Lyft, and Grab) often use the surge pricing mechanism to adjust their fare dynamically based on real-time supply and demand conditions. For example, Uber rides in 2015 frequently involved surge pricing: 15% in New York City, 17% in Los Angeles, 25% in San Francisco, and 28% in Chicago (Cohen et al. 2016). Though ride-sharing companies claim that surge pricing can connect riders with drivers more quickly,[1] customers grumble about unfairness and price gouging,[2] while drivers complain about the unpredictability of surge pricing.[3]

In the academic literature, the causal effect of surge pricing on the ride-sharing market (e.g., driver earnings and driver supply) is not well understood. Guda and Subramanian (2019) argue that surge pricing can increase market efficiency and improve social welfare in two ways. First, by pricing out riders who have lower willingness to pay, surged prices efficiently allocate the limited supply of drivers to riders who have higher willingness to pay. Second, because surge pricing increases drivers' compensation within a zone, it can increase supply by attracting more drivers to shift to that zone from adjacent zones. However, there are opposing views because surge pricing may not always benefit everyone, and it discriminates against riders with lower willingness to pay (Dholakia

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[1] Source: `https://www.uber.com/gb/en/drive/driver-app/how-surge-works/`.

[2] See `https://www.mylondon.news/lifestyle/travel/uber-bolt-londoners-complain-insane-22024015`.

[3] See `https://www.cnet.com/tech/tech-industry/detest-ubers-surge-pricing-some-drivers-dont-like-it-either/`.

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2015). Without a randomized controlled experiment, previous empirical studies have heavily relied on structural models to evaluate the impact of surge pricing. Depending on the model specifications and empirical setups, previous empirical studies have generated mixed results. For example, Ming et al. (2019) demonstrate that surge pricing improves driver welfare and platform revenue; however, Castillo (2019) shows that despite an increase in revenue during peak hours, drivers' average earnings across a shift may actually decrease, leading to decreased driver welfare. These mixed results reveal the need to better understand the causal effect of surge pricing and its implications so that firms can make informed surge pricing decisions.

In this paper, we contribute to the literature by investigating the *causal effects* of surge pricing on driver revenue and driver (labor) supply, leveraging a unique quasi-experimental setting associated with a ride-sharing platform in China. The platform introduced surge pricing in two cities at different times in 2019 – October 11 for Guangzhou (City A hereafter) and December 11 for Shenzhen (City B hereafter). The *two-month time gap* between the launch of surge pricing in these two cities provides a quasi-experimental setting that enables us to causally identify the effects of surge pricing via a difference-in-differences (DiD) design. To obtain better comparability between the treatment and control groups, we combine the DiD design with the *causal forest algorithm* that matches drivers in a nonlinear, high-dimensional space.[4] In addition to its robustness to potential selection bias, causal forest also has the ability to produce *conditional average treatment effects* for each individual driver, which allows us to further investigate how treatment effects vary across drivers.

Using the above approach, we begin by analyzing the causal effect of surge pricing on drivers' weekly revenue. Our first result indicates that at the individual level, treatment drivers on average experienced a 13.7% increase in their weekly revenue compared with their counterparts in the control city.[5] In our setting, the ride-sharing platform was charging drivers a fixed fraction of the ride fare

[4] The causal forest is a recent development in the causal machine learning literature (Athey et al. 2019, Athey and Wager 2019).

[5] At the city level, we find that the aggregate weekly revenue generated by all sampled drivers in the treatment city increased by 11.6% compared to the control city.

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as commissions, similar to Uber's and DiDi's initial practices (Bai et al. 2019). Hence, our results imply that surge pricing can increase the platform's revenue as well.

Next, we investigate two "margin factors" that could explain the increased weekly revenue induced by surge pricing. Specifically, we consider (1) intensive margin factors, i.e., drivers' labor supply patterns on a working day; and (2) extensive margin factors, i.e., drivers' decisions about the number of days to work within a week.[6] Despite the positive average treatment effect (ATE) of surge pricing on the drivers' weekly revenues, our second result reveals a significant decrease in drivers' intensive margin factors: on average, after surge pricing was launched, the treatment drivers complete fewer orders (i.e., trips) and earn less on a working day.

To gain a better understanding of our second result, we decompose the drivers' daily revenues by trip type. Besides a ride-booking service (hailing service hereinafter) similar to UberX, the focal company also offers nonstandard booking services (nonhailing service hereinafter) such as advance booking, airport pickup, and half-day/whole-day rental service. Our analysis unveils two opposing effects created by the introduction of surge pricing. On the one hand, since surge pricing only applies to hailing trips that are most likely to occur during peak hours, we uncover strong empirical evidence for drivers' switching from nonhailing trips to hailing trips, and from off-peak hailing trips to peak-hour hailing trips, focusing on hailing trips with longer travel distances. We refer to this first mechanism as the "cherry-picking effect." On the other hand, we find that the cherry-picking effect intensifies driver competition, which offsets the benefit of surge pricing; in the end, the treatment drivers did not earn more from peak-hour hailing trips. We refer to this second mechanism as the "competition effect." Ultimately, the two countervailing effects created by surge pricing jointly led to the observed decreased daily revenue on working days.

Recognizing that surge pricing leads to decreased daily revenue (our second result), the increased weekly revenue (our first result) can be attributed to drivers' adjustment of their extensive margin

[6] As established in the labor economics literature (see Heckman 1993), after a change in the system occurs, individuals can respond along (1) the "intensive margin" by varying their intensity of work on a working day, and/or (2) the "extensive margin" by deciding whether or not to work on a specific day.

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factors (in terms of the number of working days). The estimated ATE of surge pricing on drivers' extensive margin represents our third result: drivers increased the number of their working days (per week) by 12.1% on average after surge pricing was implemented, likely to compensate for the decrease in the intensive margin (i.e., drivers' daily revenues).

In addition to examining the ATE of surge pricing on drivers' revenue and labor supply, we investigate how the treatment effect varied across drivers. To do so, we use an "unsupervised clustering algorithm" to classify drivers into full-time and part-time segments. We then compare the individual treatment effects on the previously examined labor supply factors across the two segments. As our fourth result, our analyses uncover the following segment-specific effect: while surge pricing increased the size of the "pie" (in terms of total revenue), the driver pool became larger, resulting in a smaller slice per driver and therefore decreased daily revenue for all participating drivers. We find that full-time drivers were discouraged and worked on fewer days; however, part-time drivers, who were more flexible with less capacity constraint, made up for the platform-wide loss of supply by working more days per week. Overall, even though surge pricing can create more value (in terms of total revenue), our analysis reveals that the created value is unevenly distributed across drivers.

Our paper makes several contributions. From a theoretical perspective, we develop a theoretical framework for analyzing the effect of surge pricing on driver behavior based on two economic theories for characterizing the labor supply of flexible workers: the "neoclassical labor supply theory" and "reference-dependence theory". From a methodological view, to the best of our knowledge, we are the first to empirically study the causal impact of surge pricing on driver revenue and driver labor supply in a quasi-experimental setting. Further, different from previous structural model studies on surge pricing that rely on the assumption that all drivers are homogeneous, our combination of the DiD design and the causal forest method enables us to assess heterogeneous treatment effects across drivers. From a substantive point of view, we provide a more accurate characterization of the driver response to surge pricing by decomposing drivers' weekly revenue into extensive margin (drivers' decisions about the number of working days per week) and intensive margin (driver behavior on working days). In addition, our heterogeneity analyses allow us to provide managerial insights about different impacts of surge pricing on different types of drivers.

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2. Related Literature and Hypotheses Development

We provide a brief literature review before we establish our hypotheses in this section.

2.1. Literature Review

This paper is related to three streams of literature. First, it relates to the burgeoning empirical research about the effects of dynamic pricing in the ride-sharing context. From the riders' perspective, Cohen et al. (2016) leverage Uber's surge pricing algorithm to estimate demand elasticities and find that demand is quite inelastic (i.e., price elasticities vary between -0.4 and -0.6). They highlight the importance of dynamic pricing for balancing real-time demand and supply. From the drivers' perspective, Chen and Sheldon (2015) empirically examine how surge pricing affects how long a driver works on Uber's platform, conditional on having decided to start working. They find that Uber drivers drive more when earnings are high, and flexibly adjust to drive more during high surge times. Chen et al. (2019) use data on Uber drivers' hourly earnings and driving behavior to analyze the value of work flexibility in the gig economy. They find that Uber drivers earn more than twice the surplus they would in less flexible arrangements. In a case study based on Uber, Hall et al. (2015) find evidence suggesting that surge pricing can result in allocating drivers to riders with higher willingness to pay, and attracting more drivers to serve in the zone where surge pricing is in effect.

Some studies use structural models to analyze the effect of surge pricing on ride-sharing platforms via counterfactual experiments. For example, Ming et al. (2019) estimate a discrete choice model by using demand and supply data obtained from Didi Chuxing, and they use counterfactual analysis to demonstrate that surge pricing improves consumer and driver welfare as well as platform revenues. Castillo (2019) constructs a structural model of rider behavior and driver behavior and then estimates the model using data collected from Uber in Houston. He finds that surge pricing increases total welfare by 3.53% of gross revenue. The gains go mainly to riders: rider surplus increases by 6.97% of gross revenue, whereas driver surplus and platform profits decrease by 1.97% and 1.42% of gross revenue, respectively. In the absence of experimental settings, existing studies rely heavily on structural modeling assumptions and result in mixed conclusions. Different from the existing research,

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we establish the *causal effect* of surge pricing by using a unique quasi-experiment setting in which a ride-sharing company introduced surge pricing in different cities over different periods of time. In addition, we allow for heterogeneity across drivers and demonstrate that, while the average effect of surge pricing on drivers is positive, the benefit is unevenly distributed across drivers: surge pricing can be harmful to some drivers.

Second, our research also broadly relates to the theoretical research on surge pricing in operations management and economics. These studies either derive the optimal pricing policy (e.g., Bimpikis et al. 2019, Besbes et al. 2020, Özkan 2020)[7] or analyze the value of dynamic pricing from a theoretical perspective. Most studies find that surge pricing improves resource allocation and platform profit. For instance, Cachon et al. (2017) find that the optimal dynamic pricing contract substantially increases a platform's profit and that although surge pricing is not optimal, it generally achieves close to the optimal profit. Castillo et al. (2017) show that without surge pricing, drivers may have to travel a longer distance to pick up riders, causing a wild goose chase problem. Guda and Subramanian (2019) study the impact of surge pricing on supply adjustment in different market zones and demonstrate that surge pricing can provide incentives for a better balance of workers in different market zones so that the platform can earn a higher profit. Other studies find little or no effect of dynamic pricing. For example, Banerjee et al. (2015) consider the value of dynamic pricing in a model with random arrivals of consumers and providers, and find that dynamic pricing provides no benefit in terms of maximizing the platform's expected profit or system welfare. These papers typically focus on the platform's pricing decision with the assumption that drivers are fully rational. We contribute to this literature by demonstrating that drivers may not be fully rational and that there exists substantial heterogeneity across drivers in response to surge pricing, which can be useful to platforms when making pricing or driver incentive decisions.

Finally, by investigating the effect of surge pricing on driver behavior and labor supply decisions, our paper complements the behavioral operations literature (Croson et al. 2013), the service operations literature (Heineke and Davis 2007), as well as the existing literature that examines price effects

[7] Please see Yan et al. (2020) for a review.

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on the labor supply in the taxi industry and the ride-sharing industry. A number of studies have examined the effect of price on driver behavior, but with mixed results. For example, by using taxi data, Camerer et al. (1997) find a negative-wage elasticity of daily hours of work for New York City (NYC) taxi drivers, which indicates reference dependence in drivers' labor supply behavior. Crawford and Meng (2011) build on Camerer et al. (1997) and estimate a structural model of taxi drivers' decisions to stop working with more sophisticated reference-dependent preferences. Ashenfelter et al. (2010) analyze the impact of two permanent fare increases on taxi drivers and find an elasticity of -0.2 of the labor supply in response to these fare increases. At the same time, there is also evidence that price increases may motivate drivers to increase their labor supply. For example, Farber (2015) analyzes the complete record of NYC taxi trips from 2009 to 2013, and finds that drivers tend to respond positively to both unanticipated and anticipated increases in earnings opportunities. Allon et al. (2018) find that financial incentives have a significant positive influence on the decision to work and on the number of hours to work. Brodeur and Nield (2018) analyze data associated with rides in NYC provided by taxis, Lyft, and Uber. They show that there is a significant increase in the number of Uber and Lyft rides during rainy hours, whereas the increase in the number of taxi rides is much smaller, which implies that surge pricing encourages an increase in labor supply. We build on this literature on driver labor supply decisions by examining heterogeneous driver responses to surge pricing and disentangling driver responses into extensive margin and intensive margin factors. By exploring the effects of surge pricing on different aspects of driver behavior and allowing for heterogeneity across drivers, we are able to explore the mechanisms and reconcile the mixed results established in prior research.

2.2. Hypotheses Development

2.2.1. Theoretical Framework In this section, we discuss how surge pricing in the ride-sharing market may affect driver revenues and labor supply decisions following the theoretical framework as depicted in Figure 1.

When developing our hypotheses, we focus on the effects of surge pricing on drivers' revenues for two reasons. First, revenue is a direct proxy for driver welfare, which has been commonly used

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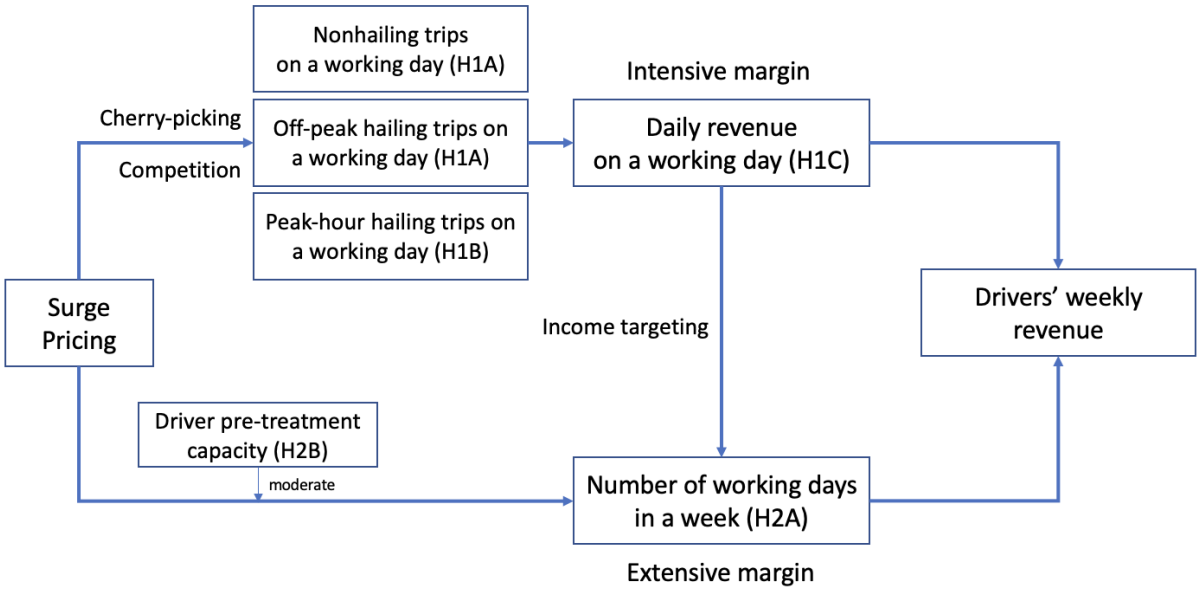

Figure 1 Effects of Surge Pricing on Driver Behavior: Theoretical Framework and Hypotheses

as a key outcome variable in previous studies on surge pricing (Lam and Liu 2017, Castillo 2019, Ming et al. 2019, Besbes et al. 2020). Second, the ride-sharing platform we study adopted a revenue-sharing business model by taking a fixed percentage of the trip fare as the commission from each ride. Therefore, any treatment effect on driver revenue also applies directly to the platform revenue. Even if a platform does not charge commissions but rather a flat platform fee, an increase in driver earnings will likely attract new drivers and mitigate existing drivers' churn rates, benefiting the platform in the long run.

We disentangle weekly revenue into two "margins" of the labor supply following the labor supply literature (see Heckman (1993)): (1) intensive margin factors, specifically, drivers' labor supply patterns on a working day; and (2) extensive margin factors, specifically, drivers' decisions about the number of days to work within a week. Further, the intensive margin factors consist of measures on three types of trips: nonhailing trips, hailing trips during off-peak hours, and hailing trips during peak hours. Finally, the effect of surge pricing on the extensive margin factors may vary across drivers with different pre-treatment capacities.

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Next, we develop our hypotheses on the effect of surge pricing on the aforementioned outcomes based on two economic theories: the "neoclassical labor supply theory" and "reference-dependence theory." Both theories have been widely used in the economics literature for characterizing the labor supply decisions of flexible workers (e.g., Camerer et al. 1997, Crawford and Meng 2011, Farber 2015, Thakral and Tô 2021), which fit well in our empirical context.

2.2.2. Effects of Surge Pricing on Drivers' Intensive Margin Factors We first derive the impact of surge pricing on drivers' intensive margin factors, i.e., daily revenue on a working day, by adapting the neoclassic labor supply theory. The original theory posits that drivers choose the number of hours to work by considering the trade-off between the utility of additional income and the disutility of additional efforts; therefore, drivers work more when wages are high and consume more leisure when wages are low, resulting in a positive correlation between wages and work hours (Lucas Jr and Rapping 1969). This is a direct result of drivers' dynamic optimization problem and implies that drivers would cherry-pick more attractive options if there are changes in the marginal utility of available options in the system (e.g., MaCurdy 1981, Farber 2015).

In our empirical context, drivers can take either hailing trips or nonhailing trips; however, surge pricing only applies to the former. From a single driver's perspective, surge pricing can increase the profitability of hailing trips, especially during peak hours. Hence, based on the neoclassic labor supply theory, we expect the driver to exhibit cherry-picking behavior by switching from nonhailing trips to hailing trips, and from off-peak hailing trips to peak-hour hailing trips. Therefore, we hypothesize that:

H1A: Due to the cherry-picking effect, surge pricing leads to a decrease in revenue generated from nonhailing trips and off-peak hailing trips for a driver during a working day.

However, the neoclassic labor supply model is based on a single driver's labor supply decision without accounting for competition among drivers in the ride-sharing market. The competition effect is an essential element arising from various online platforms. For instance, in the context of crowd-sourcing platform contests, a larger reward may attract more contestants to participate, but may

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also lead to increased competition that deters contestants from entering (Liu et al. 2021). In the context of taxi service, Buchholz (2021) develops a spatial equilibrium model and shows that expecting excess profitability in one location, taxi drivers may systematically exhibit cherry-picking behaviors by relocating to that location, which will eventually increase the competition in the location and arbitrage off the extra profitability. Following a similar logic, in our empirical context, expecting higher profitability due to surged prices, other drivers may also strategically cherry-pick peak-hour hailing trips, which will lead to intensified competition and offset the benefit of surge pricing. Therefore, we hypothesize that:

H1B: Due to the competition effect, surge pricing does not lead to an increase in revenue generated from peak-hour hailing trips for a driver during a working day.

Notice that a driver's daily revenue is comprised of revenues generated from nonhailing trips, off-peak hailing trips, and peak-hour hailing trips completed during a working day. Hence, in view of the combination of the cherry-picking effect and the competition effect, we conjecture the following net effect: surge pricing can cause the driver's daily revenue to decrease. The conjecture echoes with Castillo (2019)'s model prediction, which motivates us to establish the following hypothesis:

H1C: Surge pricing leads to a decrease in daily revenue for a driver during a working day.

2.2.3. Effects of Surge Pricing on Drivers' Extensive Margin Factors The reference-dependent preferences theory in economics posits that the utility function of an economic agent is concave relative to a certain reference value (Tversky and Kahneman 1991, Kőszegi and Rabin 2006). This theory has been used by economists to explain the negative correlation between New York taxi drivers' work hours and hourly wages (e.g., Camerer et al. 1997, Crawford and Meng 2011). These studies find that taxi drivers in New York City have a target level of daily income in mind and thus exhibit salient reference-dependent preferences: when the hourly wages are lower on a working day, drivers will work for more hours to reach the daily income target; and vice versa.

Motivated by the concept of the reference-dependent preferences theory and the income targeting behavior of taxi drivers, we establish our hypothesis about the effect of surge pricing on drivers'

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extensive margin factors (i.e., how many days to work in a week) in the ride-sharing context. The aforementioned studies mainly focus on the daily income target because most taxi drivers lease their vehicles from the taxi company on a daily basis, making it natural for them to mentally form a daily income target. However, in the ride-sharing context, there are two main reasons why a weekly income target is more appropriate than a daily income target. First, drivers typically own their cars and do not need to pay a daily rental fee. In Camerer et al. (1997)'s classic paper on reference dependency in taxi drivers' labor supply behavior, they explicitly discuss that the driver's income target depends on the lease fee payment frequency: "... fleet drivers use their daily lease fee as one reference point and are particularly reluctant to quit before reaching it (compared with lease-drivers who pay weekly or monthly, and owner-drivers)." Second, ride-sharing platforms such as Uber and Lyft often make payments to drivers on a weekly basis.[8] As such, it is natural for drivers to form a weekly income target. Therefore, the weekly income target is consistent with the ride-sharing setting as considered in our model. Based on Hypothesis H1C that surge pricing leads to decreased daily revenue, the income targeting theory implies that drivers are likely motivated to work on more days in a week in order to make up for the decreased daily revenue. Therefore, we hypothesize that:

H2A: Surge pricing leads to an increase in the number of working days within a week for a driver.

Finally, previous empirical studies of surge pricing have largely ignored heterogeneity of the treatment effects across drivers for model tractability. We fill in this gap in the research literature by exploring the heterogeneous treatment effects. Since we posit that drivers need to work on more days in a week to reach their weekly income target, we expect drivers' pre-treatment capacity to moderate the effects of surge pricing. Specifically, if a driver was a full-time driver and already working at full capacity before surge pricing was introduced, there would be very limited room for this driver to further increase her extensive margin by working more days. Therefore, we expect the treatment

[8] See, for example, `https://help.uber.com/driving-and-delivering/article/how-do-i-receive-my-earnings-?nodeId=42973e65-45a8-4aaf-90d5-d3e97ab61267`; `https://www.lyft.com/hub/posts/get-paid-instantly-with-the-lyft-direct-debit-card-for-drivers`.

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effects of surge pricing on the extensive margin to be more salient for part-time drivers who should have more flexibility in adjusting their labor supply. We hypothesize that:

H2B: The treatment effect of surge pricing on extensive margin factors is larger for part-time drivers than for full-time drivers.

In summary, to better understand the underlying mechanisms through which surge pricing affects drivers' weekly revenue, we develop our hypotheses around three effects (a *cherry-picking effect*, a *competition effect*, and an *income targeting effect*) derived from two theories (*neoclassic labor supply theory* and *reference-dependent preferences theory*). Also, we investigate four labor supply factors (the revenue generated from peak-hour hailing trips, from off-peak hailing trips, and from nonhailing trips, and the number of working days within a week) that affect a driver's weekly revenue (which is directly related to the platform's weekly revenue under the revenue sharing scheme). Next, we empirically test these hypotheses with data that we will describe in the next sections.

3. Background and Data Description

To empirically examine how surge pricing affects driver behavior, we collect detailed trip-level data from one of the largest ride-sharing platforms in China. A unique feature of our setting is that, although the platform offers a wide range of ride-sharing services to its customers, surge pricing applies to hailing trips only and not to nonhailing trips. This allows us to test our Hypothesis H1A whether drivers indeed strategically re-optimize their driving strategies and cherry-pick hailing trips that are subject to surge pricing and thus more profitable.

The focal company introduced surge pricing in City A, referred to as the "treatment" city, on October 11, 2019. Then surge pricing was introduced subsequently in City B, referred to as the "control" city, on December 11, 2019. The launch of surge pricing was communicated to both drivers and riders via in-app notifications only on the day of the launch. Therefore, there should be no pre-announcement effect that might contaminate our DiD analyses. Also, we wish to note that our analyses should not be affected by the COVID-19 pandemic because our sample period was before any COVID-19 announcement provided by the Chinese government. Specifically, the earliest official

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announcement of pneumonia of unknown etiology (i.e., COVID-19) by the Wuhan Government took place on December 31, 2019, which does not overlap with our data period.[9]

Cities A and B are geographically proximate and similar in terms of GDP, population size, and transportation infrastructure. According to a report jointly published by PwC and China Development Research Foundation, these two cities had almost identical scores for their transportation infrastructure and urban planning during our data period.[10]

The surge pricing algorithm follows the same logic as Uber's surge pricing strategy: when the demand for rides is significantly higher than the supply of drivers in a region, a surge multiplier is applied to the base fare. The magnitude of the surge multiplier depends on a series of real-time trip conditions, including weather conditions (rain, snow, etc.), traffic conditions, unexpected emergencies, and so on. If a booking request triggers surge pricing, the customer can see the base fare, the surge in the fare, and the estimated total fare for the trip on the booking app. Because rates are dynamically updated based on demand in real time, the surge multiplier can quickly change over time in the same region; in addition, surge pricing is also region-specific, so that at a given time, some regions in the city may have surge pricing while other regions do not.

3.1. Data Description

For each city (treatment City A or control City B), we randomly sample a set of drivers who received at least one ride request in 2019, and we collect data on all trips associated with each sampled driver from February 2019 to December 2019. Thus our data contain a pre-treatment period of about eight months, and a post-treatment period of about two months. We focus on drivers who were active (received at least one request) during the one-month period before surge pricing was launched.[11] The raw data before cleaning contain 7,286,491 trip requests associated with 17,400 such drivers in the two cities.

[9] Source: `http://wjw.wuhan.gov.cn/gsgg/202004/t20200430_1199576.shtml`

[10] Source: `www.pwccn.com/en/research-and-insights/chinese-cities-of-opportunities-2019-report.pdf`

[11] We note that some drivers churned before surge pricing was launched. These drivers could bias our estimates for the treatment effect of surge pricing (for example, by inflating the effect of surge pricing on driver churn) and are therefore excluded from our analyses.

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Based on the raw data, we plot the histogram of the surge multipliers across surged trips in Figure 2 and the histogram of clock hours across surged trips in Figure 3. We observe that the surge multiplier was capped at 1.35 during our data period. Also, most of the surged trips (in terms of percentage of all surged trips that occurred throughout a day) occurred during morning peak hours (7 am to 9 am) and evening peak hours (5 pm to 7 pm) throughout a day. However, the distribution of surged trips may vary by day of the week. Therefore, we further plot the percentage of surged trips by clock hour for each day of the week as shown in Figure A in Appendix A, and define day-of-the-week-specific peak hours accordingly to account for such variations across different days within a week.

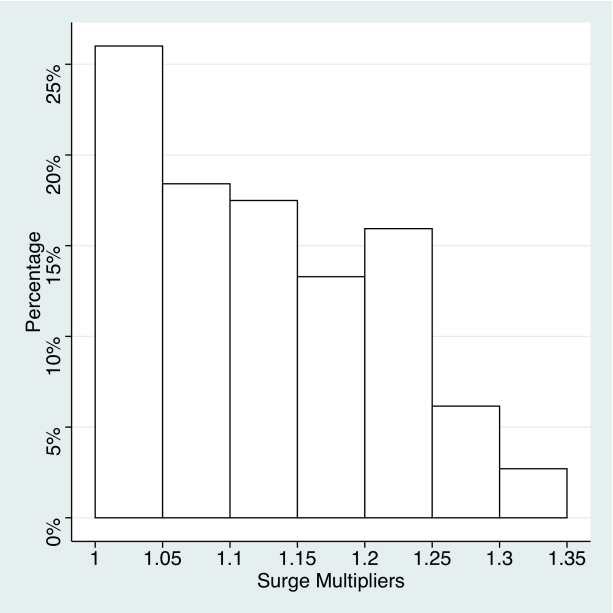

Figure 2 Histogram for Surge Multipliers

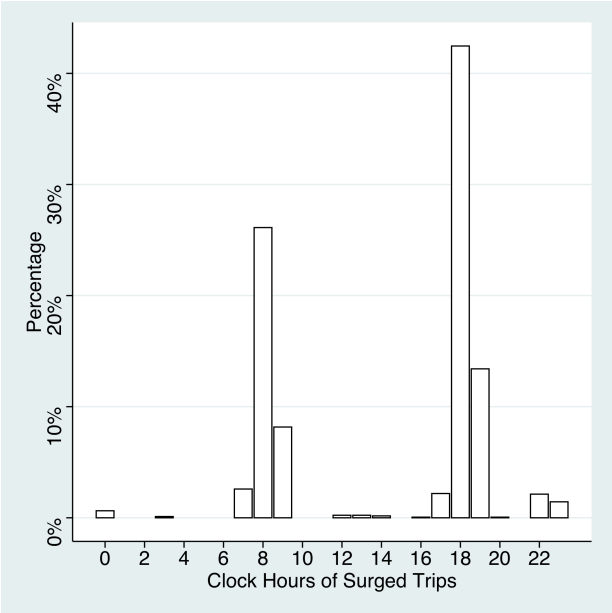

Figure 3 Histogram for Clock Hours of Surged Trips

Our data include detailed information about both "completed orders (trips)" and "unfulfilled booking requests". We consider a booking request to be a completed order (trip) if the paid price for the booking request is not a missing value in our data. There can be multiple reasons for unfulfilled booking requests, including passenger no-shows, passenger cancellations, or driver cancellations. Each booking request records information about the driver ID, the rider ID, the job type (hailing request or nonhailing request), and the request time. Hailing requests were automatically allocated by the platform to nearby drivers. For nonhailing jobs, a "request" was recorded only when a driver

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bid for a job submitted by a rider in advance. For completed trips, we further observe the driver confirmation time, the trip start and end time, the distance traveled in kilometers, the base fare, a fare adjustment due to surge pricing, and the final fare paid after surge adjustment and promotional discounts.[12] Besides transaction data, we also observe drivers' demographic information, including gender, driving tenure (number of years of driving), working tenure (number of years of working for the focal company), and whether the driver was certified as a "gold driver" by the focal company.[13]

Finally, we clean the raw data to facilitate our empirical analyses by using the steps as reported in Table 1, where each observation is either a completed trip or an unfulfilled booking request. First, as the surge pricing was subsequently introduced in the control city on December 11, 2019, we remove observations after this date for both cities so that the final sample remains unaffected by the introduction of surge pricing in the control city. Second, we follow the approach proposed by Farber (2015) for removing erroneous observations and outliers. Specifically, we remove (1) completed orders that have nonpositive travel time or last for more than 24 hours, (2) completed orders that have nonpositive distances or are longer than 100 km, and (3) booking requests that are not in the driver's home city.[14] In the end, over 6.9 million booking requests, or 95.17% of the raw data, were retained. Among these remaining booking requests, 4,431,676 (63.90%) are completed orders.

Table 2 reports the summary statistics of the cleaned data. Panel A reports the summary statistics for the 4,431,676 completed trips: the average trip was 11 km, it took 23 minutes to complete,[15] and the average price riders paid for a trip was 41 Chinese Yuan (about 6 US Dollars) after surge pricing

[12] Due to the focal company's data storage policy, we only observe the prices of completed trips but not for unfulfilled booking requests. We acknowledge this as a limitation of our data.

[13] Drivers must satisfy 15 criteria set by the focal company in order to qualify for gold driver certification. The criteria include no risky driving history, proficient knowledge of the city map, a clean and odor-free vehicle, free WiFi on board, and so on.

[14] We observe that some drivers provided trips in both cities. We define a driver's home city as the one that accounts for over 90% of the driver's total requests.

[15] The number of observations for trip duration is smaller than the number of observations for distance because the trip end time is missing for many completed trips due to the company's data storage policy.

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Table 1 Data Cleaning Steps

Step	Reason for removal	# obs removed	# obs left	% obs left
0	Raw data before cleaning		7,286,491	100.00%
1	Obs of both cities after surge pricing was introduced in control city	338,905	6,947,586	95.35%
2	Trip duration: nonpositive or longer than 24 hours	5	6,947,581	95.35%
3	Trip distance: nonpositive or longer than 100 km	12,687	6,934,894	95.17%
4	Trips in nonhome city	538	6,934,356	95.17%

adjustment and promotional discounts. Panel B reports the summary statistics for demographic variables of the 17,400 drivers: the average driver had 9.5 years of driving experience and 0.8 years of working experience with the company as of January 1, 2020,[16] almost all (99.6%) drivers were male, and 7% were certified as gold drivers.

Next, we aggregate the data into a driver-week panel from which we can compute the weekly revenue as well as the extensive margin factor, i.e., the number of working days per week. The summary statistics at the weekly level are reported in the last two panels of Table 2: Panel C reports the unconditional summary statistics (i.e., if a driver did not work in a week, then all weekly variables for that week have values of zero), and Panel D reports the summary statistics conditional on drivers working during a week. We observe that, conditional on working during a given week, the average driver earned 606 Yuan, actively worked for 4 days,[17] received 23 booking requests, and completed 15 orders within that week; drivers on average worked during 57.3% (298,603/520,586) of all weeks within our sample period.

[16] Because our causal forest analyses are conducted at the driver level, we choose January 1, 2020 as the common date for computing the time-invariant experience measures for each driver. Our results are robust to other dates because only the difference between the drivers' experience matters in our empirical setup.

[17] Because different drivers work for different numbers of hours within a day, we count a driver as actively working on a given day if he or she received at least one booking request.

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Table 2 Summary Statistics

Variables	Unit	*N*	Mean	St. Dev.	Pctl(25)	Median	Pctl(75)
A. Completed trips							
Trip distance	km	4,431,676	10.85	10.29	4.38	7.61	13.49
Trip duration	minutes	2,927,484	22.63	16.60	12.17	18.90	28.78
Original price	Yuan	4,431,676	46.64	54.39	21.00	31.61	50.89
Price paid	Yuan	4,431,676	41.09	47.21	18.00	28.00	45.02
B. Driver demographics							
Driving tenure	years	17,400	9.45	6.79	4	8	12
Working tenure	years	17,400	0.82	0.52	0.42	0.68	1.15
Male gender	indicator	17,400	0.996	0.06	1	1	1
Gold driver	indicator	17,400	0.07	0.26	0	0	0
C. Weekly level (unconditional)							
Weekly revenue	Yuan	520,586	337.54	752.71	0	0	317
Number of days working	count	520,586	2.13	2.55	0	1	4
Number of requests	count	520,586	12.88	25.08	0	1	14
Number of completed orders	count	520,586	8.24	17.10	0	0	8
D. Weekly level (conditional on working during the week)							
Weekly revenue	Yuan	298,603	606.21	919.73	64	254.8	757.3
Number of days working	count	298,603	3.86	2.23	2	4	6
Number of requests	count	298,603	23.22	29.66	3	11	31
Number of completed orders	count	298,603	14.77	20.51	2	6	19

Finally, to provide preliminary DiD comparison, we further break Table 2 into two groups (treatment/control) in two settings (pre-treatment/post-treatment) and report the summary statistics (mean and standard deviation) for each group in Table A-1 in Appendix B. The last column computes the "naive DiD estimates" (i.e., (Treatment Post-Treatment Pre)-(Control Post-Control Pre)) for each variable. We observe from Panels B and C in Table A-1 that all naive DiD estimates are positive, which seems to suggest that surge pricing could lead to higher weekly revenue, more working days, and more completed orders in a week.

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3.2. Motivating Evidence: The Effect of Surge Pricing on the Broader System

Before investigating the impact of surge pricing on individual drivers, we conduct an exploratory study of the causal impact of surge pricing on drivers' aggregate outcomes at the city-week level. To do this, we run a linear DiD regression model that is specified as follows:

$$log(Outcome_{it}) = \beta_0 + \delta Treated_i * Post_t + \theta_i + \eta_t + \varepsilon_{it}, \tag{1}$$

where $Outcome_{it}$ is the outcome variables aggregated over the sampled drivers in city i ($i = A, B$) in week t, $Treated_i$ is a binary indicator for the treatment city (i.e., City A), $Post_t$ is a binary indicator that equals 1 if week t is after the introduction of surge pricing and 0 otherwise, θ_i is the city fixed effect, and η_t is the week fixed effect. To comprehensively investigate the causal effects of surge pricing on the broader system, we examine a full set of outcome variables, including the logarithm of city-level weekly number of requests, the average order-to-request ratio (defined as the percentage of booking requests[18] that are completed, which measures the average completion rate of booking requests), the logarithm of city-level weekly number of completed orders, and the logarithm of city-level weekly revenue.

The DiD estimates are reported in Table 3. Column (1) indicates that after surge pricing was introduced, the weekly total number of requests received by the treatment drivers increased by 18.4% compared with those of the control drivers. However, the negative estimate in Column (2) indicates that surge pricing led to a decreased order-to-request ratio. As a result of the two countervailing effects, the treatment effect of surge pricing on the weekly number of completed orders (Column (3)) is positive and significant, albeit with a smaller magnitude than that in Column (1). Column (4) reveals that the total weekly revenue generated by all drivers in our sample in the treatment city (City A) increased by 11.6% after the platform introduced surge pricing. Because the platform has a fixed-commission revenue-sharing business model, the total platform revenue from the treatment

[18] For hailing jobs, requests represent real-time on-demand requests received from riders. For nonhailing jobs, because a "request" was recorded only when a driver bid for the job in advance as explained in Section 3.1, booking requests represent "matched" requests in advance.

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drivers increased as well.[19] Note that the underlying assumption of the DiD analysis is that the treatment group and the control group should have parallel trends in the pre-treatment period. We empirically verify this assumption is indeed valid in Appendix C.

Table 3 Results of the DiD analyses at the City-Week Level

	(1)	(2)	(3)	(4)
	log(number of requests)	order-to-request ratio	log(number of orders)	log(revenue)
Treated * *Post*	0.184^{***}	-0.056^{***}	0.098^{***}	0.116^{***}
	(0.000)	(0.000)	(0.000)	(0.000)
City fixed effects	Y	Y	Y	Y
Week fixed effects	Y	Y	Y	Y
Observations	88	88	88	88
R^2	0.961	0.983	0.960	0.961

Notes: This table reports the result of the DiD analysis of the city-level weekly outcomes from week 6 to week 49 in 2019. The dependent variables are the logarithm of city-level weekly number of requests (Column (1)), the average order-to-request ratio (Column (2)), the logarithm of city-level weekly number of completed orders (Column (3)), and the logarithm of city-level weekly revenue (Column (4)) from sampled drivers. *Treated* is a binary indicator for the treatment city. *Post* is a binary indicator for observations after the introduction of surge pricing. Standard errors are reported in parentheses. *** Significant at the 1% level.

Having demonstrated in Table 3 that surge pricing can increase the size of the "pie" for all sampled drivers in the treatment city, we describe the causal forest model in Section 4 and then test our proposed hypotheses in terms of the causal impact of surge pricing on individual drivers' labor supply (extensive margin and intensive margin) in Section 5. Following that, we investigate the distribution of the surplus across individual drivers with different driver-specific characteristics in Section 6.

[19] Note that our random sample of drivers only includes those who had joined the platform before surge pricing was introduced. If surge pricing attracted new drivers to join the platform, then the impact of surge pricing on the city-level revenue would be even larger. Therefore, our DiD estimate in Table 3 is a conservative estimate for the actual treatment effect.

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4. Research Design and Methodology

In this section, we discuss in detail how we combine the causal inference approach with a recent development in the area of machine learning to estimate the ATE as well as the heterogeneous treatment effects of surge pricing on individual driver behaviors.

We denote the data as (X_i, Y_i, W_i). Given a driver i, X_i is a set of covariates that characterize driver i's behavior in the pre-treatment period (i.e., before surge pricing was launched), Y_i indicates the outcome variables of interest, and W_i is a binary indicator that equals 1 if driver i was in City A, where surge pricing was implemented, and 0 otherwise. Our main objective is to estimate the causal effects of surge pricing on driver behaviors. Following the potential outcome framework (Rubin 1974), we can write the ATE conditional on $X_i = x$ as

$$\tau(x) = \mathbb{E}\left[Y_i^{(W_i=1)} - Y_i^{(W_i=0)} \mid X_i = x\right] \tag{2}$$

if the following unconfoundness assumption holds:

$$\left\{Y_i^{(W_i=0)}, Y_i^{(W_i=1)}\right\} \perp W_i \mid X_i. \tag{3}$$

Conveniently, Equation (3) can hold for a randomized controlled field experiment in which drivers are randomized into a treatment group that receives surge pricing and a control group that only receives baseline pricing. However, such an experiment can be highly costly to implement in practice, and if conducted within the same city, can be subject to the concern that there may exist spillovers from treated drivers to control drivers. Fortunately, the sequential introduction of surge pricing by our focal company in two separate yet similar cities provides us with an ideal quasi-experiment setting to study the causal impact of surge pricing without the concern of potential spillovers.

Due to their geographic proximity and economic similarities, the control city (City B) should have similar time trends to the treatment city (City A). In this case, drivers from the control city can serve the purpose of controlling for the common time trends in the driver behaviors absent surge pricing. Therefore, we can then perform a DiD analysis, comparing the behavioral changes of the treatment group drivers before and after the launch of surge pricing to the behavioral changes of

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the control group drivers during the same period. That is, in Equations (2) and (3), we take Y_i as the "change" in driver i's labor supply measures going from the pre-treatment to the post-treatment period. Such variable construction can also difference out any time-invariant driver-specific factors that may contribute to the differences in the outcome variables.

However, note that the validity of the DiD design critically relies on the parallel trend assumption, i.e., the assumption that the difference between the treatment and control drivers would be stable over time in the absence of the treatment. Such an assumption is likely to hold if treated drivers are comparable to control drivers. To achieve better comparability between the treated and control drivers, we combine the DiD design with a recent development in the causal machine learning literature, namely, the causal forest method developed by Wager and Athey (2018) and Athey and Wager (2019).

The causal forest method stems from the random forest algorithm in machine learning (Breiman 2001) and aims to provide a nonparametric approach to estimating heterogeneous treatment effects and reducing imbalance in pre-treatment covariates between the treated and control groups, thereby reducing the degree of model dependence and any nonparallel pre-trends. Similar to other causal inference techniques using observational data such as PSM (e.g., Rosenbaum and Rubin 1983, Hirano et al. 2003) and synthetic control (e.g., Abadie et al. 2010), the intuition behind the causal forest is to construct for treated drivers a set of drivers in the control group who share similar characteristics in a nonlinear, high-dimensional space during the training stage. The set of control drivers produced by the algorithm can then generate near-perfect clones for the treated drivers, such that the cloned control drivers are highly comparable to the treated drivers in the same leaves. The high comparability makes it feasible to use the cloned control drivers to compute the appropriate counterfactuals for the treated drivers.

We choose the causal forest estimator over the conventional linear regression based DiD estimator for two reasons. First, to achieve better comparability between the treatment group and control group, the conventional DiD method is often accompanied by some form of matching, which generally relies

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on parametric models to generate scores as a data preprocessing step in order to selectively prune units prior to applying a model to estimate a causal effect. For instance, in a DiD study, researchers may first estimate propensity scores using a logistic regression in the matching stage. However, due to the lack of theoretical guidance on which covariates to include and how to include them (i.e., whether to include higher-order terms or interactions) in the logistic regression, PSM can be sensitive to the exact model specification and subject to researchers' cherry-picking behavior, which often increases imbalance, inefficiency, model dependence, research discretion, and bias (King and Nielsen 2019). In contrast, the causal forest is a nonparametric but computationally efficient approach that, by virtue of not relying on parametric assumptions, is more robust to model misspecification and covariate selection. For instance, when training the causal forest, if a covariate does not significantly improve the splitting criteria, the training algorithm will not split based on this covariate. This process is data-driven and less subject to researchers' selection. In this sense, the causal forest method is more robust to covariate selection issues.

Second, previous matching methods depend on a second-stage linear regression analysis to establish statistical inferences for the ATE. To explore the heterogeneous treatment effects, researchers have to slice the sample into subgroups and re-estimate the treatment effects for each subgroup. This approach can suffer from the so-called curse of dimensionality as the covariate space increases and can even become problematic if the heterogeneous ATE is a nonlinear function of the covariates. In contrast, the causal forest can conveniently estimate heterogeneous treatment effects for each individual, and the estimates are consistent and asymptotically normal when the covariate space is large (Athey et al. 2019).

Owing to these two advantages, we use the causal forest algorithm to estimate the causal impact of surge pricing. Similar methods have been adopted by several recent studies (Cui and Davis 2019, Iyengar et al. 2020, Pattabhiramaiah et al. 2021, Guo et al. 2021). We provide a review of the causal forest algorithm, how we construct the outcome variables, and how we implement the causal forest with a DiD design in our empirical setting in Appendix D.

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5. Results: Average Treatment Effects

By combining the DiD design with the causal forest method, we obtain our estimates for the ATEs of surge pricing on drivers' weekly revenue and other key labor supply factors, as depicted in Figure 1. We first describe the result for the ATE on drivers' weekly revenue. We then test our hypotheses proposed in Section 2.2 by decomposing the ATE on weekly revenue into intensive and extensive margin factors. Note that the underlying identifying assumption of causal forest is the overlap assumption (Athey et al. 2019). In Appendix F, we show that this assumption holds in our empirical exercise, and the covariates are well balanced across the treatment group and control group after applying causal forest.

5.1. Average Treatment Effect of Surge Pricing: Drivers' Weekly Revenue

We begin by investigating the ATE of surge pricing on individual drivers' weekly revenue. The impact on revenue not only directly links to participating drivers' welfare, but also links to the ride-sharing platform's revenue under the revenue-sharing scheme. When estimating the ATE, we construct a doubly robust average treatment effect estimator by augmenting the naive plugin estimator with a residual-based correction (Chernozhukov et al. 2018).

The point estimate, standard error, and 95% confidence interval for the ATE on drivers' weekly revenues are reported in Table 4. The point estimate is positive, with a magnitude of 0.137, and this estimate is statistically significant at the 1% level. Note that when constructing the dependent variable, we use logarithms, and therefore the interpretation of the point estimate should be in relative terms – that is, compared with control group drivers in City B, the treated drivers in City A on average increased their weekly revenue by 13.7% after the surge pricing was introduced in City A. This result suggests that after the launch of surge pricing, drivers were better off in terms of their weekly revenues, which seems to contradict the finding of Castillo (2019) that driver welfare decreases due to surge pricing.

To gain a better understanding of this result, we next decompose the weekly revenue into intensive and extensive margin factors to test our proposed hypotheses in Section 2.2 and investigate the underlying mechanisms that explain the increased weekly revenue.

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Table 4 Results for Average Treatment Effects on Weekly Revenue

Variables	ATE	Std. Dev.	95% Confidence Interval
Weekly revenue	0.137***	0.046	[0.048, 0.226]

Notes: *** Significant at the 1% level.

5.2. Average Treatment Effect of Surge Pricing: Intensive Margin Factors

In this section, we test our proposed hypotheses related to the effects of surge pricing on drivers' intensive margin factors, i.e., daily revenue on a working day and its composition.

5.2.1. Cherry-Picking Effect As we articulated in Section 2.2, surge pricing increases the expected profitability of hailing trips relative to nonhailing trips (i.e., trips that are not subject to surge pricing), especially during peak hours. Therefore, based on the neoclassic labor supply theory, drivers may switch to hailing trips by accepting fewer nonhailing trips and may shift their work schedule from off-peak hours to peak hours with the hope to complete more peak-hour hailing trips with surge pricing, leading to decreased revenue from nonhailing trips and off-peak hailing trips (as stated in Hypothesis H1A). To test this hypothesis, we examine the ATE of surge pricing on the daily number of requests, daily number of completed orders, daily revenue generated from nonhailing trips and off-peak hailing trips in Panels A and B of Table 5, respectively.

First, we observe from Panel A that the daily number of nonhailing requests (i.e., those nonhailing requests accepted by drivers in advance as explained in Section 3.1), the daily number of completed nonhailing orders, and the daily revenue generated from nonhailing trips decreased after surge pricing was introduced. Since surge pricing does not affect the price of nonhailing trips, riders' demand for nonhailing trips should be unaffected by surge pricing; therefore, the observed decreased number of nonhailing requests, number of orders, and revenue of nonhailing trips should mainly be driven by drivers' decisions rather than riders' behavior: treated drivers decided to switch away from nonhailing trips to hailing trips, since the expected profitability from the latter was higher due to surge pricing.

Second, Panel B indicates that the daily number of completed orders and revenue for off-peak hailing trips also decreased significantly after surge pricing was introduced. Following a similar logic,

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Table 5 The Mechanisms of the Effects of Surge Pricing

Variables	ATE	Std. Dev.	95% Confidence Interval
A. Nonhailing Trips			
daily number of requests	-0.029***	0.009	[-0.046, -0.011]
daily number of orders	-0.018***	0.007	[-0.032, -0.005]
daily revenue	-0.037***	0.014	[-0.063, -0.010]
average order-to-request ratio	0.034*	0.020	[-0.006, 0.074]
B. Off-peak Hailing Trips			
daily number of requests	0.031	0.024	[-0.017, 0.078]
daily number of orders	-0.092***	0.027	[-0.145, -0.038]
daily revenue	-0.122***	0.047	[-0.214, -0.030]
average order-to-request ratio	-0.034***	0.005	[-0.044, -0.025]
C. Peak-hour Hailing Trips			
daily number of requests	0.097***	0.027	[0.045, 0.150]
daily number of orders	-0.032	0.022	[-0.076, 0.012]
daily revenue	-0.048	0.038	[-0.123, 0.028]
average order-to-request ratio	-0.048***	0.006	[-0.059, -0.036]

Notes: * Significant at the 10% level. *** Significant at the 1% level.

this pattern is driven by drivers' cherry-picking behavior in shifting their work schedule from non-peak to peak hours, since surge pricing is more likely to be triggered during peak hours. The above results document the existence of a cherry-picking effect of treated drivers gravitating toward peak-hour hailing trips and hence lend strong support for Hypothesis H1A.[20]

5.2.2. Competition Effect. Table 5 reveals the presence of a cherry-picking effect induced by surge pricing. As we posit in Section 2.2, in a competitive market, such cherry-picking behavior would likely result in increased competition among drivers as more drivers gravitate towards hailing trips during peak hours, and this competition effect may offset the benefit of surge pricing. To identify the net effect of surge pricing, we estimate and report the ATE of surge pricing on peak-hour

[20] Besides drivers' switching towards peak-hour hailing trips, we also find that surge pricing led to drivers cherry-picking hailing trips with longer distances (please see Table A-3 in Appendix E).

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hailing trips in Panel C of Table 5. Interestingly, the estimates reveal that for peak-hour hailing trips, despite an increase in the daily number of requests,[21] the daily number of completed hailing trips during peak hours and daily revenue obtained by treated drivers during peak hours did not correspondingly increase, which suggests that the competition effect has largely offset the cherry-picking effect. Consequently, the treated drivers did not earn more from hailing trips during peak hours on working days, supporting Hypothesis H1B.

To empirically verify that competition is intensified during peak hours, we test whether the total number of active drivers during peak hours in the treatment city indeed increased relative to the control city after surge pricing was introduced. Specifically, we run a linear DiD regression analysis at the city-hour level by estimating the following model:

$$ln(N_{it}) = \beta_0 + \delta\, Treated_i * Post_t + \theta_i + \eta_t + \varepsilon_{it}, \tag{4}$$

where $ln(N_{it})$ is the logarithm of the number of active drivers working in city i during hour t,[22] and we focus on peak hours as defined in Section 3.1. $Treated_i$ is a binary indicator that equals 1 if city i is the treatment city and 0 otherwise. $Post_t$ is a binary indicator that equals 1 if the observation was taken after the introduction of surge pricing and 0 otherwise, θ_i is the city fixed effect, and η_t is the day-hour fixed effect. The key parameter of interest is δ, which measures the ATE of surge pricing on the number of drivers at the city-hour level. Specifically, δ would be positive if surge pricing did entice more drivers to work during peak hours, intensifying the competition effect.

The estimated coefficients for Equation (4) are reported in Column (1) of Table 6. The hourly number of drivers during peak hours indeed increased by 20.2% in the treatment city compared with the control city after surge pricing was introduced, which again manifests increased competition for peak-hour hailing trips resulting from surge pricing. Columns (2) and (3) replace the dependent

[21] Riders may strategically request hailing trips repeatedly, hoping to get nonsurged prices during peak hours, which would lead to the observed increase in the number of requests.

[22] The number of active drivers in city i during hour t is defined as the count of distinct drivers in city i who received at least one booking request during hour t.

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variable with the logarithm of respectively the number of requests and the number of orders per driver. The results are consistent with our previous results in Table 7: despite an increase in the average number of requests per driver, the average number of completed orders per driver decreased by 8%, largely due to the competition effect. (We shall discuss Columns (4) and (5) later.)

Table 6 Evidence for the Competition Effect During Peak Hours

	(1) log(# drivers)	(2) log(# requests/# drivers)	(3) log(# orders/# drivers)	(4) log(# full-time drivers)	(5) log(# part-time drivers)
*Treated * Post*	0.202***	0.010***	−0.080***	0.165***	0.309***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
City FE	Y	Y	Y	Y	Y
Day-hour FE	Y	Y	Y	Y	Y
N	3,432	3,432	3,432	3,432	3,432
R^2	0.961	0.795	0.946	0.965	0.909

Notes: This table reports the results for analyses of the peak-hour competition effect at the city-hour level as specified in Equation (4). The dependent variables are the logarithms of, respectively, the count of drivers, the number of requests per driver, the number of completed orders per driver, the count of full-time drivers, and the count of part-time drivers, during each peak hour on each day. *Treated* is a binary indicator for the treatment city. *Post* is a binary indicator for observations after the introduction of surge pricing. All columns control for fixed effects for city and day-hours. Robust standard errors are clustered at the city level in parentheses. *** Significant at the 1% level.

5.2.3. Daily Revenue. Recognizing these two countervailing effects, we now examine the ATE of surge pricing on the *daily total distance driven*, *daily number of completed orders*, and *daily revenue*, as reported in Table 7. Observe that the point estimates for the key intensive margin factors are all negative and statistically significant at the 1% level. This observation implies that, after surge pricing was implemented, treated drivers in City A ended up driving shorter distances, completing fewer orders, and earning lower daily revenues on a working day, which supports Hypothesis H1C.

This finding signifies the importance of our decomposition exercise: despite the increased weekly revenue, we show that treated drivers in fact earned less and were worse off on working days after surge

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Table 7 Results for ATE on Intensive Margin Factors

Variables	ATE	Std. Dev.	95% Confidence Interval
Daily total distance driven	-0.138***	0.040	[-0.216, -0.061]
Daily number of completed orders	-0.123***	0.027	[-0.176, -0.069]
Daily revenue	-0.178***	0.047	[-0.270, -0.087]

Notes: *** Significant at the 1% level.

pricing was introduced. Our analyses also provide important managerial implications for ride-sharing platforms. In a highly competitive ride-sharing market where drivers are strategic, introducing surge pricing can give rise to two countervailing effects: a cherry-picking effect and a competition effect. If there is no centralized system coordinating drivers' labor supply decisions, the competition effect can dominate the cherry-picking effect, and consequently, surge pricing may backfire and unexpectedly decrease drivers' daily revenue on working days. To alleviate such issues, the ride-sharing platform may benefit from coordinating drivers on their work schedules and search strategies, for instance, by providing drivers with a real-time heat map of regional labor supply. With better information on the spatial distribution of competing drivers in the market, drivers can better adjust their work strategies to mitigate overcompetition.

5.3. Average Treatment Effect of Surge Pricing: Extensive Margin Factors

We have verified our hypotheses that treated drivers' daily revenue on working days decreased after surge pricing was introduced; therefore, the increased weekly revenue can only be attributed to the adjustment in treated drivers' extensive margin factors. As discussed in Section 2.2, based on the reference-dependent preferences theory, drivers may have a weekly income target. Therefore, when there is a decrease in the daily revenue, we expect drivers to work on more days to reach their mental target on weekly income.

In Panel A of Table 8, we report the point estimate, standard error, and 95% confidence interval for the ATE on treated drivers' number of working days per week. We find that the ATE of surge pricing is positive with a magnitude of 0.121 and statistically significant at the 1% level. Therefore, after

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the introduction of surge pricing, the treated drivers indeed increased their extensive margin of labor supply by working more days per week to compensate for the decreased daily revenue, supporting Hypothesis H2A.

We conclude the section by investigating how drivers allocated the increased extensive margin across weekday/weekend and day/night shifts. Specifically, we categorize completed orders into 4 groups, including weekday day orders, weekday night orders, weekend day orders, and weekend night orders.[23] We then estimate the ATE on the fraction of each group and report the results in Panel B of Table 8. We find that after surge pricing was introduced, drivers did not change their labor supply for day shifts on weekdays, probably because these drivers mostly work on a part-time basis (as we will show in Section 6.2, over 73.4% of drivers in our sample are part-time drivers) and have little flexibility to increase day shifts on weekdays. Meanwhile, the ATE on the fraction of weekday night shifts is positive and statistically significant, which indicates drivers allocated the increased extensive margin towards night shifts on weekdays in response to surge pricing. As for weekend work schedules, we observe a reduced fraction of weekend day orders while the fraction of weekend night orders was unaffected. In summary, the evidence in this section suggests that drivers worked on more days to make up for the decreased daily revenue, and the extensive margin increase was mainly attributed to weekday night shifts.

6. Conditional Average Treatment Effect: Heterogeneity Across Drivers

Recognizing the possibility of driver heterogeneity, we now extend our analysis by exploring how the Average Treatment Effect (ATE) varies across drivers. To do so, we leverage a unique feature of the causal forest method to "retrieve" *individual* conditional average treatment effects (CATEs). Specifically, we estimate the CATE of surge pricing on the key outcome variables for each individual driver by using "out-of-bag predictions".

In this section, we first present the summary statistics for the estimated individual CATEs to confirm that heterogeneity across drivers is indeed present. We then use a clustering analysis to

[23] Day orders are defined as completed orders with trip start time between 6 am and 6 pm.

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Table 8 Results for ATE on Extensive Margin Factors and Work Schedules

Variables	ATE	Std. Dev.	95% Confidence Interval
A. Extensive Margin			
Weekly number of working days	0.121***	0.027	[0.067, 0.174]
B. Work Schedules			
Fraction of weekday day orders	-0.004	0.006	[-0.015, 0.007]
Fraction of weekday night orders	0.011**	0.005	[0.001, 0.021]
Fraction of weekend day orders	-0.008*	0.004	[-0.017, 0.000]
Fraction of weekend night orders	0.001	0.004	[-0.006, 0.009]

Notes: * Significant at the 10% level. ** Significant at the 5% level. *** Significant at the 1% level.

investigate how the increased weekly revenues (due to surge pricing) are distributed across driver segments. The purpose of our heterogeneity analyses is to help company managers and policymakers better understand how the treatment effects of surge pricing vary across drivers so that they can develop more effective policies targeting specific groups of drivers.

6.1. Empirical Evidence of Heterogeneous Treatment Effects

Based on the out-of-bag predictions, Table 9 presents the statistics for the estimated *individual* CATE on six key driver supply measures: (a) weekly revenue, (b) daily revenue on a working day, (c) number of working days per week, (d) daily revenue from peak-hour hailing trips, (e) daily revenue from off-peak hour hailing trips, and (f) daily revenue from nonhailing trips. In addition, Figure 4 depicts the empirical distributions of these measures, with the red dashed vertical line in each histogram representing the corresponding arithmetic mean of the CATE. We do observe significant heterogeneity of individual treatment effects across drivers. Take the CATE on weekly revenue as an example. First, the magnitude of individual CATEs has a wide range, from a minimum of -1.266 to a maximum of 1.328; the standard deviation, 0.282, is more than twice that of the mean, 0.137, suggesting large dispersion in the CATE on weekly revenue across drivers. Second, from the histogram of individual CATEs (Figure 4(a)), we observe that a considerable fraction of drivers fall into the negative range and supposedly became worse off due to the surge pricing, despite the overall

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positive ATE as highlighted by the red dashed vertical line. In addition to the weekly revenue, the remaining labor supply factors all exhibit pronounced heterogeneity across drivers. Such pronounced heterogeneity across drivers motivates us to further explore how individual treatment effects vary across driver segments and characteristics.

Table 9 Summary Statistics for Estimated Individual Treatment Effects

Variable	Mean	Std. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Weekly revenue	0.137	0.282	−1.266	0.030	0.187	0.308	1.328
Daily revenue	−0.180	0.203	−1.215	−0.263	−0.162	−0.077	0.634
Weekly number of working days	0.120	0.170	−0.693	0.045	0.147	0.233	0.769
Daily revenue: peak-hour hailing	−0.051	0.189	−1.056	−0.127	−0.060	0.021	1.017
Daily revenue: off-peak hailing	−0.121	0.187	−1.242	−0.186	−0.100	−0.015	0.733
Daily revenue: nonhailing	−0.037	0.081	−0.635	−0.079	−0.042	0.001	0.377

6.2. Heterogeneous Effects of Surge Pricing: Full-time Versus Part-time Drivers

In Section 5.2, we showed that the positive treatment effect of surge pricing on treated drivers' weekly revenue was largely driven by the increase in the extensive margin of drivers' labor supply (i.e., working more days). A driver who was already working at full capacity would have very limited room to further increase the extensive margin (by working more days). Therefore, we expect drivers with different pre-treatment capacities to respond differently to surge pricing.

We proceed with a K-means clustering algorithm (Hartigan and Wong 1979) to cluster drivers into segments for each city based on their labor supply measures in the pre-treatment period. In practice, the K-means clustering algorithm can easily run into "curse of dimensionality" problems. That is, as the number of covariates for clustering increases, the distance-based similarity measure can converge to a constant value, making the clustering algorithm less effective at distinguishing between drivers. To avoid such a problem, we select two most important variables that sufficiently characterize drivers' pre-treatment capacity as the clustering covariates: drivers' average weekly number of working days

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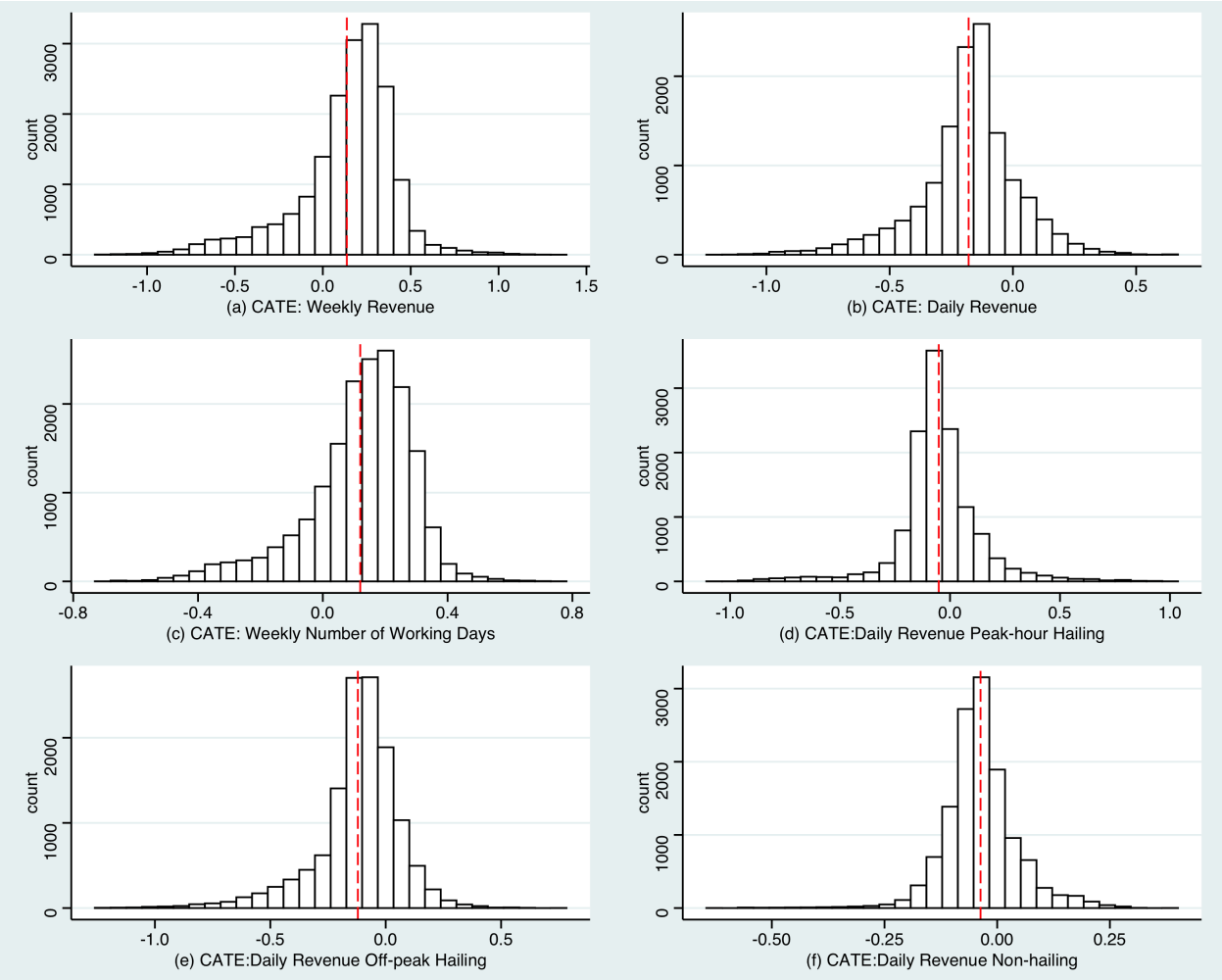

Figure 4 Histograms of the CATE on Key Labor Supply Factors across All Drivers

and average weekly number of completed orders in the pre-treatment period, both of which are normalized to have the same scales. We choose these two variables because they capture both the extensive margin and the intensive margin and, as shown below, can effectively identify distinct driver segments. Finally, the optimal number of driver clusters is determined with the average silhouette score following prior literature (e.g., Pagell et al. 2015, Roels and Staats 2021). This metric justifies a two-cluster solution for each city (please see Figure A-4 in Appendix G).

We visualize the clustering outcomes in Figure 5 and report the cluster-specific summary statistics for key pre-treatment labor supply measures in Table 10. The two selected covariates can indeed sufficiently screen out two distinct segments of drivers: we observe that in the pre-treatment period, drivers in Cluster 1 (red dots), who account for 73.4% of all drivers, on average worked for 1.41 days

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Table 10 Summary Statistics for Pre-treatment Driver Behaviors by Driver Segments

Variables	*N*	Mean	St. Dev.	Pctl(25)	Median	Pctl(75)
A. Segment 1: Part-Time Drivers						
Weekly revenue	12,763	115.91	203.08	18.70	63.22	160.38
Weekly number of working days	12,763	1.41	1.12	0.43	1.14	2.21
Daily revenue	12,763	71.30	76.69	38.71	59.15	87.37
Daily revenue: peak-hour hailing	12,763	19.28	20.33	5.74	14.62	26.00
Daily revenue: off-peak hailing	12,763	47.34	34.72	25.67	41.46	61.49
Daily revenue: nonhailing	12,763	4.68	62.80	0	0	0
B. Segment 2: Full-Time Drivers						
Weekly revenue	4,637	1,034.07	832.16	489.68	767.29	1,337.95
Weekly number of working days	4,637	5.09	0.97	4.38	5.09	5.85
Daily revenue	4,637	195.99	138.21	104.32	159.70	257.60
Daily revenue: peak-hour hailing	4,637	55.58	36.76	27.15	44.79	77.31
Daily revenue: off-peak hailing	4,637	120.98	67.19	70.42	103.21	158.32
Daily revenue: nonhailing	4,637	19.42	98.16	0.00	1.28	6.09

and earned 115.91 Chinese Yuan per week. In contrast, drivers in Cluster 2 (blue triangles), who account for 26.6% of the total drivers, on average worked for 5.09 days and earned 1,034.07 Chinese Yuan per week. For ease of reference, we label Cluster 1 as the "part-time segment" and Cluster 2 as the "full-time segment." We conduct a t-test to examine whether the CATEs on drivers' key labor supply factors are statistically different across the two segments. Columns (1) and (2) of Table 11 report the arithmetic means of CATEs for the full-time drivers and part-time drivers respectively, along with the t-test statistics on whether each mean is statistically different from 0 in parentheses. Column (3) reports the difference in the means of CATEs between the full-time drivers and part-time drivers, along with the t-test statistics on whether the difference is statistically different from 0 in parentheses.

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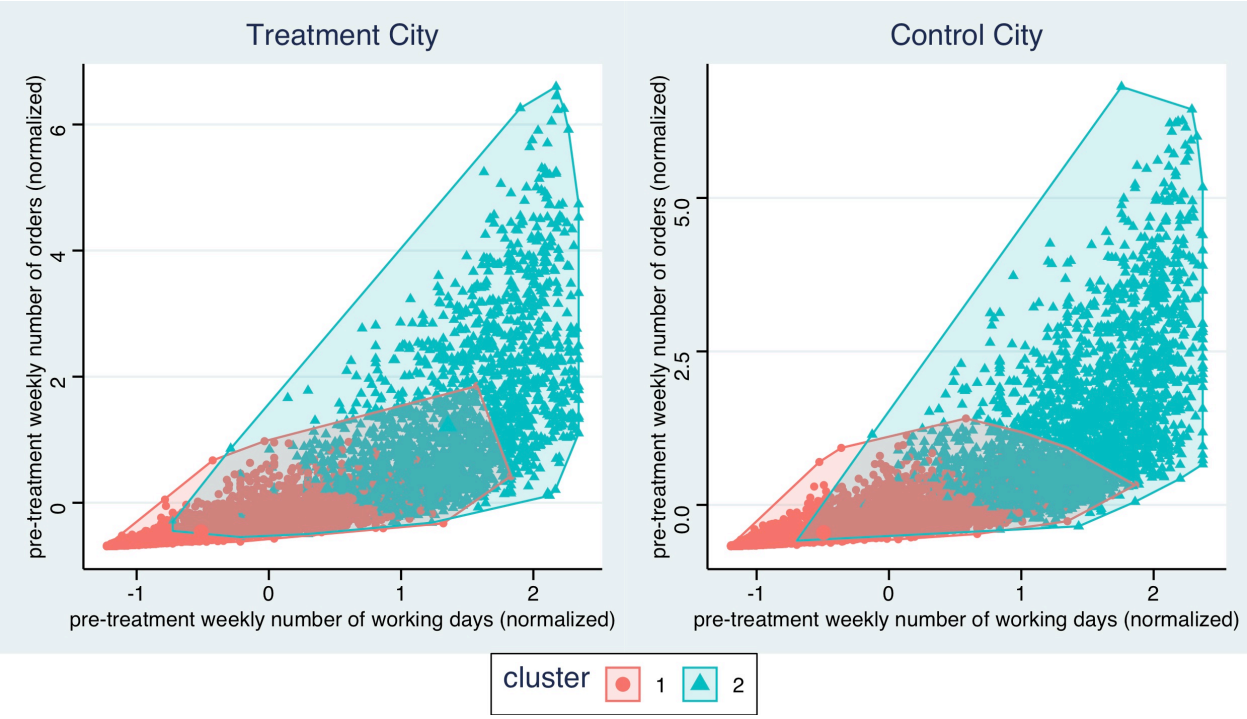

Figure 5 Visualization of the K-means Clustering Outcome by City

Row (1) of Table 11 indicates that, despite the overall positive ATE on weekly revenue, full-time drivers have a negative average CATE, while part-time drivers have a positive average CATE. In other words, the increased weekly revenue we documented in Section 5 should be primarily driven by part-time drivers, and as a matter of fact, full-time drivers were hurt by surge pricing. The results in rows (2) and (3) further unveil the reasons for the differential average CATEs on weekly revenue between the two segments.

First, row (2) shows that the positive ATE on the extensive margin of labor supply documented in Section 5 mainly comes from part-time drivers, because they had the flexibility to work on more days per week, while in contrast, full-time drivers were unable to further extend their working days because they had already been working at full capacity. Therefore, the increased capacity of part-time treated drivers intensified the competition, which may have discouraged full-time treated drivers from working. This competition between driver segments is further supported by Columns (5) and (6) of Table 6, which are based on linear DiD analyses at the city-hour level using the specification in Equation (4) and reveal a considerable larger treatment effect of surge pricing on the number

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Table 11 Heterogeneity of Treatment Effects across Driver Segments

Dependent variables	(1) Full-time Drivers	(2) Part-time Drivers	(3) Difference in Mean
(1) Weekly revenue	-0.079***	0.213***	0.292***
	(-14.559)	(132.519)	(51.566)
(2) Weekly number of working days	-0.022***	0.171***	0.193***
	(-7.049)	(170.164)	(59.659)
(3) Daily revenue	-0.114***	-0.211***	-0.097***
	(-51.684)	(-89.372)	(-30.104)
(4) Daily number of orders from nonhailing	-0.033***	-0.011***	0.022***
	(-39.609)	(-31.343)	(24.448)
(5) Daily revenue from nonhailing	-0.064***	-0.024***	0.04***
	(-40.183)	(-34.439)	(23.333)
(6) Daily number of orders from non-peak hailing	-0.077***	-0.098***	-0.021***
	(-57.255)	(-78.421)	(-11.183)
(7) Daily revenue from non-peak hailing	-0.085***	-0.138***	-0.053***
	(-39.41)	(-62.951)	(-16.962)
(8) Daily number of orders from peak-hour hailing	-0.045***	-0.029***	0.016***
	(-29.782)	(-24.275)	(8.21)
(9) Daily revenue from peak-hour hailing	-0.057***	-0.048***	0.009***
	(-22.539)	(-22.587)	(2.752)

Notes: This table reports the group means and the t-test results of the estimated CATEs for full-time and part-time drivers. The t-test statistics on whether the respective value is statistically different from zero are reported in parentheses. *** Significant at the 1% level.

of part-time drivers (30.9%) than on the number of full-time drivers (16.5%). This complementary evidence further verifies that the competition effect is primarily driven by part-time drivers flooding into the market.

Second, row (3) compares the difference in the average CATE on daily revenue: both part-time and full-time drivers suffered a sharp decrease in daily revenue due to the competition effect, but part-time drivers seem to have been adversely affected to a larger extent, possibly because they are less experienced.

The remaining rows in Table 11 examine the average CATE between the driver segments by decomposing intensive margin factors by trip type, similar to the decomposition exercise in Section

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5.2. Rows (4) and (5) reveal that due to the cherry-picking effect, the daily number of completed orders and revenue from nonhailing trips decreased, with a larger effect on full-time drivers, because nonhailing trips were mostly provided by full-time drivers in the pre-treatment period. Rows (6) and (7) indicate that, due to the cherry-picking effect, the daily number of completed orders and revenue from off-peak hailing trips also decreased, with a larger effect on part-time drivers. This is plausibly because part-time drivers tend to have more flexibility than full-time drivers for shifting their work schedules to peak hours. Rows (8) and (9) show that both part-time and full-time drivers had a slight decrease in the daily number of completed orders and revenue from peak-hour hailing trips, due to the competition effect.[24]

In summary, the above analyses show that at the individual level, part-time (full-time) treated drivers increased (decreased) their labor supply and obtained higher (lower) weekly revenue compared with similar drivers in the control city.

Impact of Surge Pricing on Driver Labor Supply. By combining our heterogeneity analyses presented in Table 11 with Columns (4) and (5) of Table 6, we obtain a more granular characterization of how surge pricing affected different drivers: surge pricing enticed more drivers to participate, particularly part-time drivers (as shown in Columns (4) and (5) of Table 6). As a result, competition among drivers intensified, resulting in revenue loss on given working days. As shown in Table 11, full-time drivers were discouraged and worked on fewer days, but part-time drivers made up for their loss by working on more days. To conclude, while the "pie" got larger (as shown in Table 3) and the driver pool got even larger, the slice per driver did not increase. In the end, even though surge pricing created more value, the increased wealth was distributed unevenly across full-time and part-time drivers. Our results highlight the importance of accounting for individual heterogeneity when analyzing the effect of surge pricing, which has not been explored in most structural analyses of surge pricing in the literature (e.g., Lam and Liu 2017, Ming et al. 2019).

[24] The estimates in Table 11 are the means of estimated *individual* treatment effects using out-of-the-bag prediction from the causal forest algorithm. Meanwhile, estimates in Table 7 are the doubly robust estimates of the *average* treatment effects. Therefore, there are differences in the estimates.

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7. Discussion and Conclusion

In this section, we discuss the theoretical and managerial implications of our study, and conclude with limitations and future research directions.

7.1. Theoretic Implications

In the absence of a randomized experimental setting, previous empirical studies have mainly relied on structural models to investigate how drivers may have behaved with and without surge pricing. However, the counterfactual results based on structural models can be highly sensitive to the underlying model assumptions (Chintagunta et al. 2006). As a result, previous studies have generated mixed conclusions regarding the causal impact of surge pricing (e.g., Lam and Liu 2017, Castillo 2019, Ming et al. 2019). To gain a better understanding of the causal effect of surge pricing, we leverage a quasi-experiment to analyze the causal impact of surge pricing on drivers' labor supply patterns following the theoretical framework put forward in Figure 1. We used a difference-in-differences design along with the causal forest method, which affords clean identification with minimal reliance on parametric assumptions and allows for heterogeneous treatment effects across drivers.

We find that, on average, surge pricing increased drivers' weekly revenue. By decomposing the weekly revenue into drivers' intensive and extensive margins, we identify two countervailing effects that are at play based on the neoclassic labor supply theory and reference-dependent preferences theory: a cherry-picking effect (H1A) and a competition effect (H1B). On average, the latter dominates, resulting in decreased daily revenue (H1C). Consequently, driven by the income targeting effect, drivers worked more days to make up for the declined daily revenues (H2A).

By demonstrating the income targeting effect in the ride-sharing market, we make an important theoretical contribution to the literature on flexible workers' labor supply decisions. There has been a heated debate in the economics literature regarding whether taxi drivers exhibit neoclassic labor supply behavior or income targeting behavior. These studies typically consider income target at the daily level because taxi drivers lease the vehicle on a daily basis. However, our results suggest that drivers' income target in a sharing (gig) economy can be formed over a longer horizon than the daily

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level: we empirically show that drivers in our data focus on a weekly income target instead of a daily one, plausibly because (1) they own the vehicles and do not need to pay a daily rent, and (2) they typically receive payouts from ride-sharing platforms on a weekly basis. This finding is theoretically interesting, highly relevant for policy making, and novel for the design of trip fares in the ride-sharing market.

Further, we find that the treatment effects of surge pricing are heterogeneous between two driver segments: surge pricing enticed more part-time drivers to flood the market and crowd out full-time drivers, implying that the increased weekly revenue was primarily driven by part-time drivers (H2B). We also show that full-time drivers who were more "loyal" to the company benefited less from or were even hurt by the introduction of surge pricing. By exploring the effects of surge pricing on different aspects of driver behavior and allowing for heterogeneity across drivers, we are able to explore the mechanisms and reconcile the mixed results established in prior research.

7.2. Managerial Implications

Our results provide important managerial implications for ride-sharing platforms regarding evaluating the impact of surge pricing in an unbiased manner. First and foremost, previous research suggests that driver earnings during a single shift may decrease, which seems to hinder ride-sharing platforms from adopting surge pricing. However, we find that drivers managed to compensate for the decline in daily revenue by switching to more profitable trips (H1A, H1B) and working more days (H2A) and, as a result, earn higher weekly revenues. Our result highlights the importance of a more granular examination of drivers' accumulative income over a period of time instead of on a single day; otherwise, the platform may not be able to properly evaluate the welfare impact of surge pricing.

Second, we show that in a decentralized ride-sharing market, an important reason for the surge-price-induced decline in daily revenue (H1C) is intensified driver competition (H1B). To mitigate this issue, the ride-sharing platform can consider providing drivers with better information on the supply density, for instance, a real-time heat map of available drivers in each zone of the city. Knowing the number of drivers available in each zone, drivers can make more informed decisions when choosing their geographical zones in order to fulfill more profitable orders.

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Third, we document drivers' cherry-picking behavior by switching from nonhailing to hailing trips (H1A) and from short-distance to long-distance hailing trips. Such cherry-picking behaviors may hurt the welfare of certain riders. For instance, riders who need nonhailing services may find it harder to get a ride. The ride-sharing platform can potentially compensate drivers on nonhailing services in order to improve rider satisfaction.

Finally, the heterogeneity analyses reveal that surge pricing might hurt full-time drivers who are more "loyal" to the platform (H2B). Therefore, in order to reduce drivers' churn rates and multi-homing behavior, the platform may need to subsidize such "loyal" drivers – for example, by lowering their commission rate or offering additional bonuses.

Our findings also provide an important implication to full-time drivers who heavily rely on providing ride-sharing services to make a living: after the introduction of surge pricing, they should expect an increasing number of part-time drivers to enter the market, resulting in decreased weekly revenue for themselves. These full-time drivers should strategically adjust their working plans accordingly. For instance, to circumvent intensified competition, they may refrain from cherry-picking peak-hour hailing trips, or relocate to less competitive regions to search for passengers.

7.3. Limitations and Future Research

There are several limitations to the present study that represent opportunities for future research. First, we focus on the causal effect of surge pricing on driver behavior because our data consist of a random sample of drivers and we observe the full trip history of these sampled drivers. While acknowledging that riders are also important players in the ride-sharing market, we are unable to examine the causal effects on rider behavior (for instance, how a rider may switch between hailing and nonhailing trips), because we do not have a similar panel dataset of riders and because riders typically multi-home across various platforms. Should rider data on multiple platforms be available, it would be a promising direction for future research to investigate the causal impact of surge pricing on rider behavior.

Second, because we do not have access to drivers' GPS trajectory data, we are not able to analyze the effect of surge pricing on drivers' spatial search strategy when examining driving efficiency.

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Should such data become available, it would be fruitful to study how drivers learn and adapt their search strategies following the introduction of surge pricing. Third, we only observe the prices of completed trips but not prices of unfulfilled booking requests, and therefore are unable to analyze drivers' response to prices in real time. Finally, because our identification relies upon a difference-in-differences design, the fact that City B introduced surge pricing only two months after City A means that we are unable to assess the long-term effects of surge pricing. Future research can investigate these long-term effects and their dynamics over time.

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Appendix

A. Histogram of Surged Trips by Day of the Week

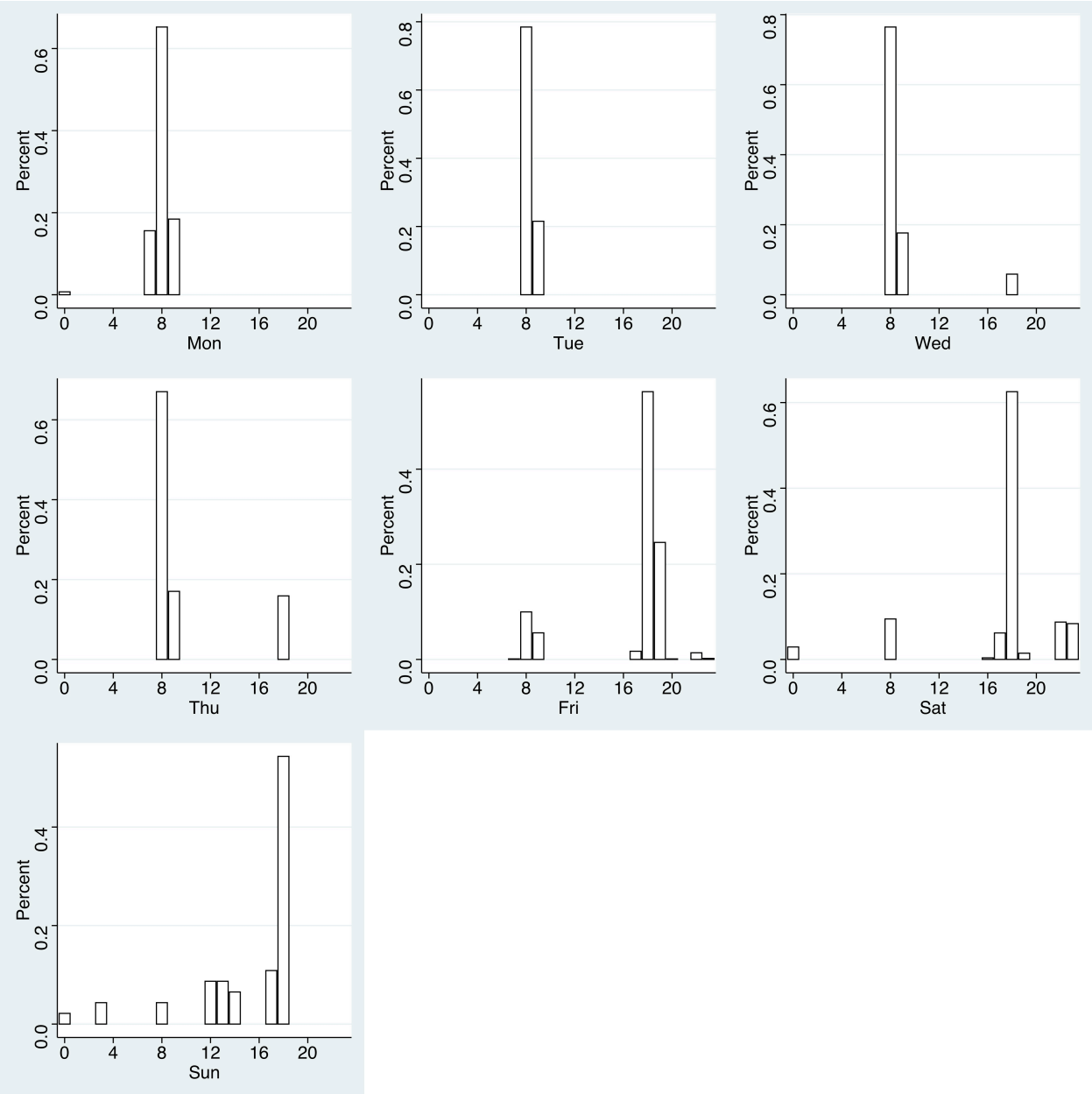

Figure A-1 Histogram of Clock Hours of Surged Trips by Day of the Week

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B. Summary Statistics by Group

Table A-1 Summary Statistics by City (Treatment and Control) and Treatment Period (Pre and Post)

Variables	Treatment City				Control City				Naive DiD
	Pre		Post		Pre		Post		
A. Completed trips									
Trip distance	11.3	(11.52)	12.5	(12.52)	10.43	(9.24)	10.65	(9.64)	0.98
Trip duration	23.21	(18.95)	24.48	(18.14)	22.07	(15.02)	22.7	(15.73)	0.64
Original price	48.86	(50.28)	54.1	(55.36)	44.33	(53.89)	47.15	(68.01)	2.42
Price paid	42.65	(45.12)	48.39	(49.88)	39	(46.07)	42.77	(57.21)	1.97
B. Weekly level (unconditional)									
Weekly revenue	357.7	(732.46)	192.41	(518.04)	410.76	(849.84)	206.07	(620.3)	39.40
Number of days working	2.27	(2.57)	1.67	(2.37)	2.36	(2.6)	1.55	(2.34)	0.21
Number of requests	13.28	(24.53)	8.68	(19.68)	15.2	(27.5)	8.9	(21.86)	1.70
Number of completed orders	8.41	(16.43)	3.96	(9.65)	10.56	(20.05)	4.81	(12.36)	1.30
C. Weekly level (conditional)									
Weekly revenue	615.87	(872.83)	407.66	(681.68)	693.5	(1009.64)	469.03	(858.36)	16.26
Number of days working	3.91	(2.22)	3.61	(2.21)	3.98	(2.22)	3.61	(2.24)	0.07
Number of requests	22.84	(28.51)	18.46	(24.74)	25.64	(31.7)	20.39	(28.61)	0.87
Number of completed orders	14.47	(19.37)	8.46	(12.45)	17.82	(23.39)	10.96	(16.35)	0.85

C. Parallel Pre-trend Tests for the DiD Analyses at the City-Week Level

An underlying assumption of the DiD analysis is that the treatment group and the control group have parallel trends in the pre-treatment period. To justify the validity of this assumption, we empirically test whether the outcome variables present differential time trends across the treated and control groups before surge pricing was introduced. Specifically, we follow Patel et al. (2021) and estimate the linear time trend across our outcome variables as follows:

$$log(Outcome_{it}) = \beta_0 + \phi_1 t + \phi_2 t^2 + \lambda_1 \, Treated_i * t + \lambda_2 \, Treated_i * t^2 + \theta_i + \varepsilon_{it}, \tag{A-1}$$

where t is the continuous pre-treatment week, $Treated_i$ is a binary indicator for the treatment city, and θ_i is the city fixed effect. To account for the non-linear effects, we include both the linear and the quadratic term of time. The estimation results are reported in Table A-2 in the Appendix. Both λ_1 and λ_2 are statistically insignificant, which indicates that there is no difference in the pre-trends in the pre-treatment period between the treatment group and control group.

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Table A-2 Results of Parallel Pre-trend Tests for the DiD analyses at the City-Week Level

	(1)	(2)	(3)	(4)
	log(number of requests)	order-to-request ratio	log(number of order)	log(revenue)
t	0.090^{***}	0.0003	0.103^{***}	0.080^{***}
	(0.013)	(0.003)	(0.014)	(0.010)
t^2	-0.001^{***}	-0.0002^{***}	-0.002^{***}	-0.001^{***}
	(0.0003)	(0.0001)	(0.0003)	(0.0002)
$Treated * t$	0.0003	−0.002	0.007	−0.012
	(0.018)	(0.004)	(0.019)	(0.014)
$Treated * t^2$	0.0003	−0.00000	0.0001	0.0004
	(0.0004)	(0.0001)	(0.0004)	(0.0003)
City fixed effects	Y	Y	Y	Y
Observations	70	70	70	70
R^2	0.903	0.930	0.874	0.900

Notes: This table reports the results of the parallel pre-trend tests (Equation (A-1)) for the DiD analyses at the city-week level. The dependent variables are the logarithm of city-level weekly number of requests, the order-to-request ratio, the logarithm of city-level weekly number of completed orders, and the logarithm of city-level weekly revenue from sampled drivers. *Treated* is a binary indicator for the treatment city. t is the continuous pre-treatment week. Standard errors are reported in parentheses. *** Significant at the 1% level.

D. Implementation of the Causal Forest

In this appendix, we first present a review of the causal forest algorithm and then discuss how we implemented a causal forest with a difference-in-differences design in our empirical setting. For more technical details about causal forests, we refer readers to Wager and Athey (2018) and Athey et al. (2019).

The causal forest method involves a training stage and an estimation stage. In the training stage, we grow B causal trees that constitute the causal forest, where each tree is grown on a random subsample of drivers and a random subset of covariates for training. Each causal tree then uses recursive partitioning, similar to Breiman (2001), that repetitively selects a splitting variable from the random subset of covariates and categorizes drivers with similar pre-treatment characteristics into the same terminal leaf until no further improvement can be achieved. Unlike the random forest algorithm, where each splitting aims to maximize information gain, the partitioning of causal trees is implemented via optimizing over a criterion (Athey et al.

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2019) that aims to increase the heterogeneity of the treatment effects across the child nodes as efficiently as possible.

After growing the causal forest, for a test-point driver with covariate x, we can then construct the weight of driver i in the training set, $\alpha_i(x)$, by counting the frequency of i falling into the same leaf as x across the B trees:

$$\alpha_i(x) = \frac{1}{B} \sum_{b=1}^{B} \frac{\mathbb{1}\left(X_i \in L_b(x)\right)}{|L_b(x)|}, \tag{A-2}$$

where $L_b(x)$ denotes the set of training set drivers falling in the same leaf as x in the bth tree, $\mathbb{1}\left(X_i \in L_b(x)\right)$ is a binary indicator of whether driver i falls in $L_b(x)$, and $|L_b(x)|$ measures the size of leaf $L_b(x)$, i.e., the total number of drivers in $L_b(x)$. At a high level, the data-adaptive weight $\alpha_i(x)$ captures how similar driver i in the training set is to a test set driver with covariate x.

Let $e(x) = \mathbb{P}[W_i \mid X_i = x]$ denote the propensity of driver i to receive the treatment W_i and $m(x) = \mathbb{E}[Y_i \mid X_i = x]$ the expected outcome marginalizing over treatment. Together with $\alpha_i(x)$, the data-adaptive kernel that measures how often driver i falls in the same leaf as the test point x, we can then follow Chernozhukov et al. (2018) to compute the doubly robust estimates of the conditional average treatment effect $\hat{\tau}(x)$ as follows:

$$\hat{\tau}(x) = \frac{\sum_{i=1}^{n} \alpha_i(x) \left(Y_i - \hat{m}^{(-i)}\left(X_i\right)\right) \left(W_i - \hat{e}^{(-i)}\left(X_i\right)\right)}{\sum_{i=1}^{n} \alpha_i(x) \left(W_i - \hat{e}^{(-i)}\left(X_i\right)\right)^2}, \tag{A-3}$$

where $\hat{m}^{(-i)}(X_i)$ and $\hat{e}^{(-i)}(X_i)$ are the out-of-bag estimators using separate regression forests for $m(x) = \mathbb{E}[Y_i \mid X_i = x]$ and $e(x) = \mathbb{P}[W_i \mid X_i = x]$. Note that, to improve the consistency of the treatment effect estimates, we incorporate "honesty" when training causal trees (Wager and Athey 2018). Specifically, when growing a causal tree b, half of the randomly selected drivers are used for recursive partitioning and the other half are used to estimate the treatment effects with Equation (A-3). The two halves of the drivers in the trees are mutually exclusive, which therefore mitigates the possibility of correlation between the training and estimation stages.

In our empirical context, the outcome variable Y_i in Equation (A-3) corresponds to the changes from the pre-treatment period to the post-treatment period in the drivers' weekly revenue and the four labor supply measures. When constructing drivers' revenue and extensive margin measures, we aggregate the trip-level data into a weekly panel. Specifically, we follow previous studies (e.g., Iyengar et al. 2020, Guo et al.

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2021) and take $Y_i^{revenue} = \frac{1}{|T_1|}\sum_t log(Revenue_{it|t\in T_1}) - \frac{1}{|T_0|}\sum_t log(Revenue_{it|t\in T_0})$ as the outcome variable of revenue for driver i, where T_0 stands for the weeks in the pre-treatment period, T_1 stands for the weeks in the post-treatment period, $|\,T_0\,|$ and $|\,T_1\,|$ respectively represent the number of weeks in the pre- and post-treatment periods, and $log(Revenue_{it})$ is the log-transformed weekly revenue of driver i in week t.[25] Such a variable construction can difference out any time-invariant driver-specific factors that may contribute to the differences in the outcome variables. Using logarithms allows us to evaluate differences in the outcome variables across drivers in relative terms.

Similarly, we construct the outcome variable for the extensive margin of labor supply as $Y_i^{extensive} = \frac{1}{|T_1|}\sum_t log(Extensive_{it|t\in T_1}) - \frac{1}{|T_0|}\sum_t log(Extensive_{it|t\in T_0})$, where $log(Extensive_{it})$ is the log-transformed weekly number of working days for driver i in week t.

The remaining intensive margin measures, i.e., the daily number of requests, number of orders, and revenues from peak-hour hailing trips/off-peak hailing trips/nonhailing trips on a working day, are constructed at the daily level. Specifically, we compute $Y_i^{intensive} = \frac{1}{|D_1|}\sum_t log(Intensive_{it|t\in D_1}) - \frac{1}{|D_0|}\sum_t log(Intensive_{it|t\in D_0})$, where D_0 stands for driver i's working days in the pre-treatment period, D_1 stands for the working days in the post-treatment period, $|\,D_0\,|$ and $|\,D_1\,|$ represent the number of working days in the pre- and post-treatment periods, respectively, and $log(Intensive_{it})$ is the logarithm of driver i's intensive margin measures on day t.

Finally, we apply two sets of covariates X_i to train the causal forests in order to find clusters of demographically comparable drivers who had similar labor supply patterns prior to the launch of surge pricing. The first set of weekly covariates characterize drivers' behaviors during the pre-treatment period, including the weekly total revenue/requests/finished orders, weekly revenue/requests/finished orders from hailing bookings, weekly revenue/requests/finished orders during peak hours, and weekly number of working days. The second set of covariates includes driver demographics, such as gender, working tenure, driving tenure, and the gold driver certification status. In total, 354 covariates are used for the causal forest matching, including 350 driver labor supply measures for each week from week 6 to week 40 (10 measures $\times$ 35 weeks), plus the four driver demographic variables.

[25] As some part-time drivers may not work during all weeks, we add a small number 0.01 within the logarithm for all outcome measures in our analyses.

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E. ATE on Trip Distance

We estimate the ATE of surge pricing on the average distance per trip for each trip type. While the average distance of nonhailing trips was largely unaffected (the estimate for average trip distance in Panel A is statistically insignificant), the ATEs of surge pricing on the average distance for peak-hour and off-peak hailing trips are both positive and statistically significant, which suggests that drivers became pickier and gravitating towards hailing trips with longer distances.

Table A-3 ATE on Trip Distance

Variables	ATE	Std. Dev.	95% Confidence Interval
A. Nonhailing Trips			
average trip distance	-0.048	0.049	[-0.143, 0.047]
B. Off-peak Hailing Trips			
average trip distance	0.095***	0.010	[0.075, 0.115]
C. Peak-hour Hailing Trips			
average trip distance	0.069***	0.012	[0.045, 0.094]

Notes: *** Significant at the 1% level.

F. Evaluating the Identifying Assumptions of Causal Forest

An underlying assumption of causal forest is the overlap assumption, which requires a positive probability of treatment for each individual driver. Statistically, it means that we should not be able to deterministically decide the treatment assignment of any individual driver conditional on covariates X_i (Athey et al. 2019). Two common diagnostics to evaluate if the overlap assumption holds are the covariate balance plot and the propensity score histogram, which we present in this section.

First, we check whether the covariates are well balanced across the treatment and control groups of drivers in our analysis by plotting the inverse-propensity weighted kernel density plots for four key pre-treatment demographic variables in Figure A-2, including drivers' pre-treatment average weekly revenue, average weekly revenue from hailing trips, average weekly revenue from peak-hour trips, and average weekly number of working days. We find that all variables are well balanced. Due to limited space, we report the kernel density plot for the four most important pre-treatment covariates. The remaining 350 covariates also yield good post-matching balance (suppressed due to limited space and available upon request).

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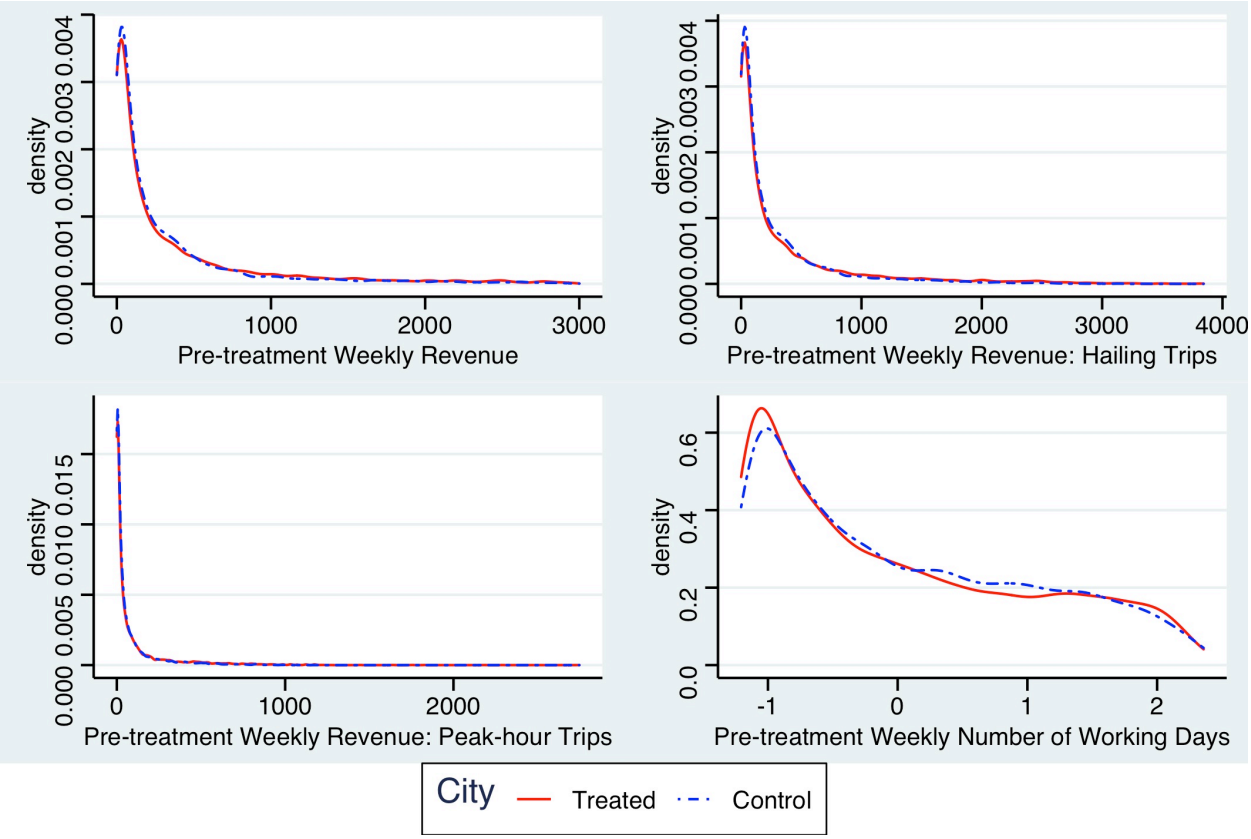

Figure A-2 Post-Matching Kernel Density Plots for Treated and Control Drivers

Second, the overlap assumption requires that none of the estimated propensity scores for individual drivers should be close to one or zero. We plot the histograms of the estimated propensity scores for both extensive margin analyses and intensive margin analyses in Figure A-3. We find that the estimated propensity scores, $\hat{W}$, have a bell-shaped empirical distribution with values tapered between 0 and 1. This pattern suggests the two cities, on average, are comprised of similar individual drivers in terms of their covariates.

In summary, the evidence presented above verifies that the overlap assumption for causal forest holds in our empirical exercise.

G. Deciding the Optimal Number of Clusters for the Clustering Analyses

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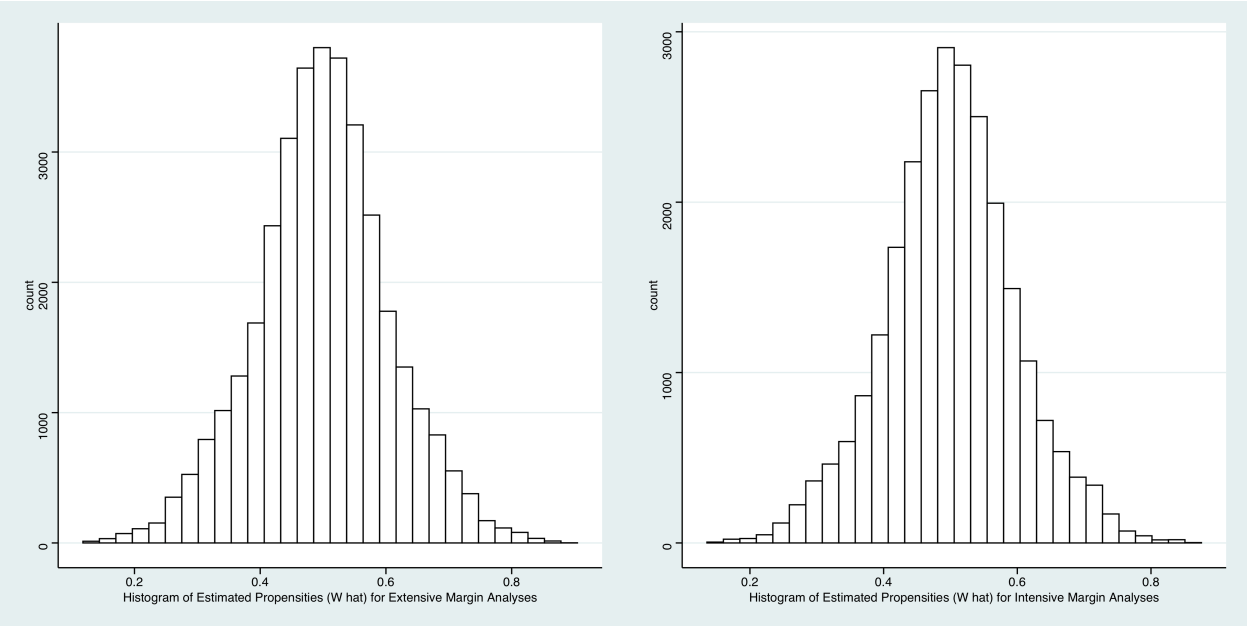

Figure A-3 Histogram of the Estimated Propensity Scores

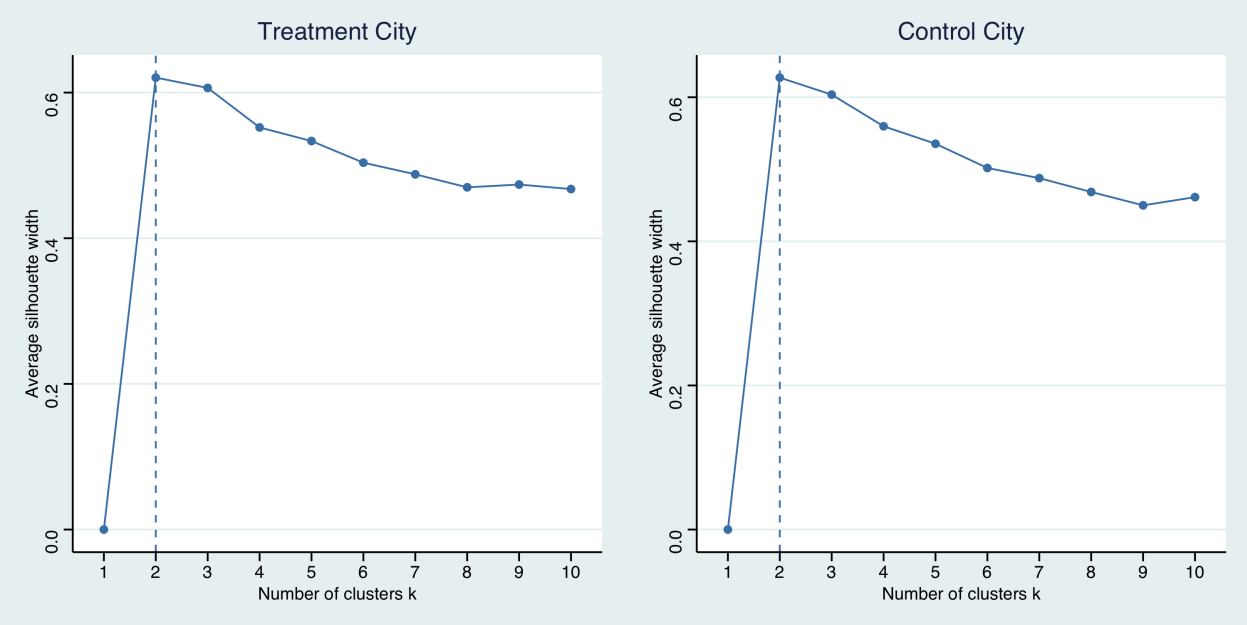

Figure A-4 The Silhouette Scores for the K-means Clustering

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